MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CONVERTER, ANALOG/DIGITAL, DIGITAL, KEY CODE QQQ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic analog/digital converter module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3 3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.70 failures per million operating hours when predicted in accordance with MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be QQQ.

3.5.2 Size. The module size shall be 1B.

3.5.3 Weight. The module shall have a maximum weight of 80 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment B.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/232-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltages of +15 Vdc ±5%, -15 Vdc ±5%, and +5 Vdc ±10%.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

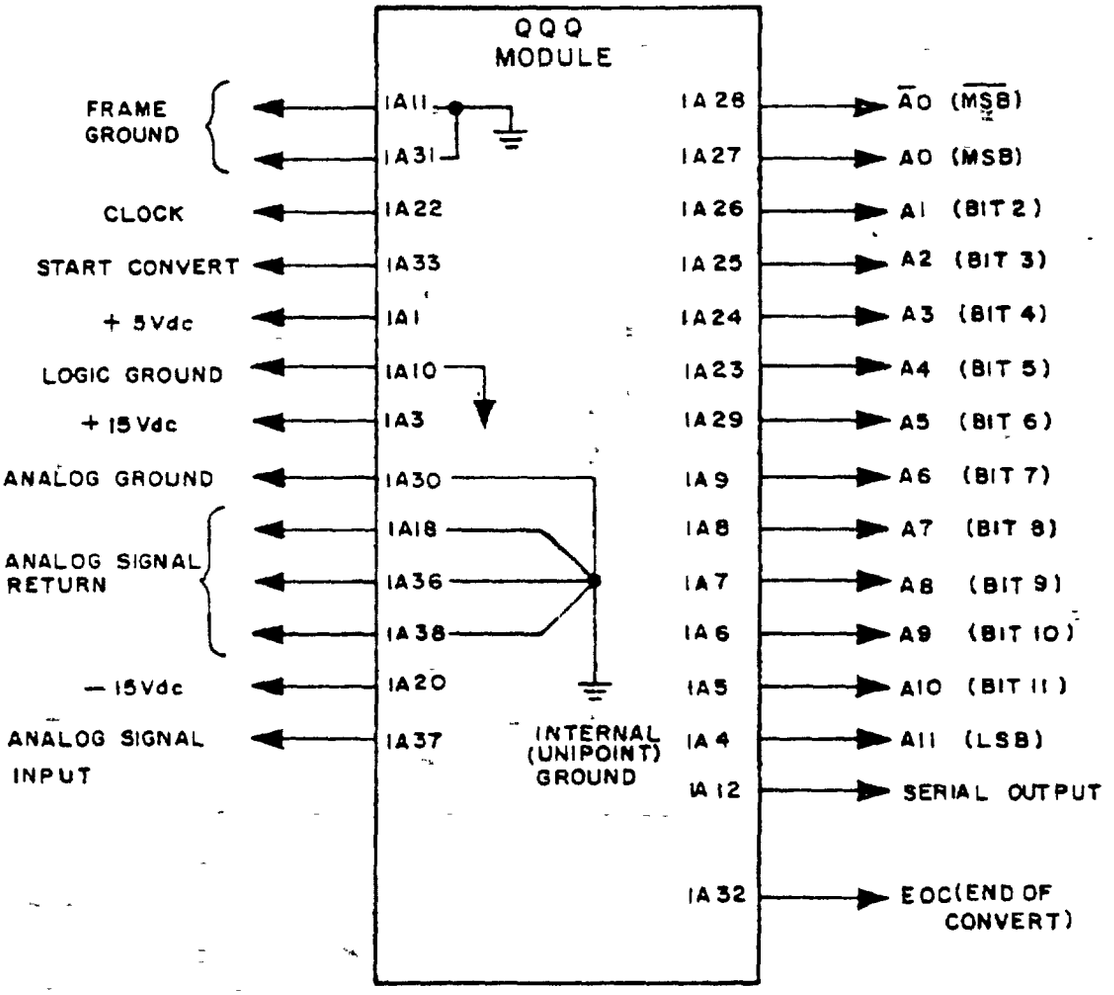
4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Z shall be performed in any order with the following exceptions. If test conditions A, B, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, F where applicable, followed by the remaining tests in the sequence. If test condition D is performed, all tests marked with an "X" in the test type column shall be performed as proof of no damage.



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NOTE: Unused pins 1A2,1A13,1A14,1A15,1A16,1A17,1A19,1A21,1A34,1A35,1A39, and 1A40.

FIGURE 1. QQQ module function diagram, analog/digital converter



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TABLE I. Module electrical requirements.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25° ±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| A1-A5 | Isolation | III | X | 10.0 | ∞ | 1.0 | ∞ | 10.0 | ∞ | 1.0 | ∞ | MΩ |
| B1-B4 | Continuity | IV | X | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | Ω |
| B5 | Continuity, frame | IV | S | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | Ω |
| B6-B8 | Continuity | IV | Q | 0.0 | 0.1 | 0.0 | 0.1 | 0.0 | 0.1 | 0.0 | 0.1 | Ω |
| C | Isolation, digital ground to analog ground | V | X | 1 | ∞ | 1 | ∞ | 1 | ∞ | 1 | ∞ | kΩ |
| D1-D10 | Power supply sequencing | VI | Q | No measurements are required. Module shall pass the remainder of this specification as proof of no damage. | | | | | | | | s |
| E1,E2 | Input breakdown current | VII | S | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | mA |
| F1-F15 | Output short circuit current | VIII | S | 0 | -25 | 0 | -25 | 0 | -25 | 0 | -25 | mA |
| G1, G2 | Transition time ($t_{TLH}$), $C_L$ = 50 pF | IX | S | 5 | 100 | 5 | 100 | 5 | 100 | 5 | 100 | ns |
| G3-G15 | Transition time ($t_{TLH}$), $C_L$ = 50 pF | IX | S | 5 | 60 | 5 | 60 | 5 | 60 | 5 | 60 | ns |
| G16, G17 | Transition time ($t_{TLH}$), $C_L$ = 150 pF | IX | Q | 10 | 200 | 10 | 200 | 10 | 200 | 10 | 200 | ns |
| G18-G30 | Transition time ($t_{TLH}$), $C_L$ = 150 pF | IX | Q | 10 | 100 | 10 | 100 | 10 | 100 | 10 | 100 | ns |
| G31, G32 | Transition time ($t_{THL}$), $C_L$ = 50 pF | IX | S | 2 | 50 | 2 | 50 | 2 | 50 | 2 | 50 | ns |
| G33-G45 | Transition time ($t_{THL}$), $C_L$ = 50 pF | IX | S | 2 | 30 | 2 | 30 | 2 | 30 | 2 | 30 | ns |

See footnotes at end of table.





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TABLE I. Module electrical requirements - Continued

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25° ±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| G46, G47 | Transition time ($t_{THL}$), $C_L$ = 150 pF | IX | Q | 5 | 100 | 5 | 100 | 5 | 100 | 5 | 100 | ns |
| G48-G60 | Transition time ($t_{THL}$), $C_L$ = 150 pF | IX | Q | 5 | 60 | 5 | 60 | 5 | 60 | 5 | 60 | ns |
| G61, G62 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | IX | S | 20 | 200 | 20 | 200 | 20 | 200 | 20 | 200 | ns |
| G63-G86 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | IX | S | 20 | 160 | 20 | 160 | 20 | 160 | 20 | 160 | ns |
| G87, G88 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | IX | Q | 20 | 220 | 20 | 220 | 20 | 220 | 20 | 220 | ns |
| G89-G112 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | IX | Q | 20 | 180 | 20 | 180 | 20 | 180 | 20 | 180 | ns |
| G113,G114 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | IX | S | 20 | 200 | 20 | 200 | 20 | 200 | 20 | 200 | ns |
| G115-G127 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | IX | S | 20 | 160 | 20 | 160 | 20 | 160 | 20 | 160 | ns |
| G128,G129 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | IX | Q | 20 | 220 | 20 | 220 | 20 | 220 | 20 | 220 | ns |
| G130-G142 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | IX | Q | 20 | 180 | 20 | 180 | 20 | 180 | 20 | 180 | ns |
| H1 | Power supply current (5.5 Vdc) | X | X | 1 | 75 | 1 | 75 | 1 | 75 | 1 | 75 | mA |

See footnotes at end of table.





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TABLE I. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25° ±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| H2-H3 | Power supply current (±15.75 Vdc) | X | X | 1 | 50 | 1 | 50 | 1 | 50 | 1 | 50 | mA |
| J1, J2 | Static input current, high level | XI | X | 0 | 20 | 0 | 20 | 0 | 20 | 0 | 20 | µA |
| J3, J4 | Static input current, low level | XI | X | -0.05 | -0.44 | -0.05 | -0.44 | -0.05 | -0.44 | -0.05 | -0.44 | mA |
| K1,K17-K30 | Static output voltage, high level | XII | X | 2.4 | 4.5 | 2.4 | 4.5 | 2.4 | 4.5 | 2.4 | 4.5 | V |
| K2-K16 | Static output voltage, low level | XII | X | 0.0 | 0.3 | 0.0 | 0.3 | 0.0 | 0.3 | 0.0 | 0.3 | V |
| L1 | Input impedance check (measuring input current) | XIII | S | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | mA |
| L2 | Input impedance check (measuring input current) | XIII | S | 0 | -1 | 0 | -1 | 0 | -1 | 0 | -1 | mA |
| M1, M7 | Input capacitance | XIV | Q | 2 | 15 | 2 | 15 | 2 | 15 | 2 | 15 | pF |
| N1-N4 | Digital input threshold and pulse width check | XV | X | ← Shall meet the requirements of table XV. → | | | | | | | | |
| P1, P2 | Start convert pulse width check | XVI | S | ← Shall meet the requirements of table XVI. → | | | | | | | | |
| Q1-Q3 | Power supply noise rejection check | XVII | Q | ← Shall meet the requirements of table XVII. → | | | | | | | | |
| R1, R2 | Serial output check | XVIII | Q | ← Shall meet the requirements of table XVIII → | | | | | | | | |
| S1-S4 | Multiplexed input delay check | XIX | Q | ← Shall meet the requirements of table XIX. → | | | | | | | | |

See footnotes at end of table.





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* TABLE I. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25° +5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| T1-T4 | Serial/parallel output check | XX | X | ← Shall meet the requirements of table XX. → | | | | | | | | |
| U1-U3, U5-U7 | Hysteresis | XXI | Q | -2.44 | 2 44 | -2.44 | 2.44 | -2.44 | 2.44 | -2.44 | 2 44 | mV |
| U4 | Hysteresis | XXI | Q | -4.88 | 4 88 | -4.88 | 4.88 | -4.88 | 4.88 | -4.88 | 4.88 | mV |
| V1, V3-V14, V16-V26 | Differential nonlinearity | XXII | X | 0 50 | 9.76 | 0.50 | 9.76 | 0.50 | 9.76 | 0.50 | 9.76 | mV |
| V2, V15 | Differential nonlinearity | XXII | X | 0.50 | 9 76 | 0 50 | 11.00 | 0.50 | 9.76 | 0.50 | 12.20 | mV |
| W1 | Linearity error | XXIII | X | 7.4976 | 7 5024 | 7.4951 | 7.5049 | 7.4976 | 7.5024 | 7.4951 | 7.5049 | V |
| W2 | | XXIII | X | 4.9976 | 5.0024 | 4.9951 | 5.0049 | 4.9976 | 5.0024 | 4.9951 | 5 0049 | V |
| W3 | | XXIII | X | 2.4976 | 2.5024 | 2 4951 | 2.5049 | 2.4976 | 2.5024 | 2.4951 | 2.5049 | V |
| W4 | | XXIII | X | -2.4 | 2 4 | -4.9 | 4 9 | -2.4 | 2.4 | -4.9 | 4.9 | mV |
| W5 | | XXIII | X | -2.4976 | -2 5024 | -2.4951 | -2.5049 | -2.4976 | -2 5024 | -2.4951 | -2.5049 | V |
| W6 | | XXIII | X | -4.9976 | -5.0024 | -4.9951 | -5 0049 | -4.9976 | -5 0024 | -4.9951 | -5.0049 | V |
| W7 | | XXIII | X | -7.4976 | -7.5024 | -7.4951 | -7.5049 | -7.4976 | -7.5024 | -7.4951 | -7.5049 | V |
| X1-X4 | Zero error | XXIV | X | -4 88 | 4.88 | -4.88 | 4.88 | -4 88 | 4.88 | -4 88 | 4.88 | mV |
| Y1-Y4 | Absolute error | XXV | X | 9.9776 | 10.0176 | 9.9776 | 10.0176 | 9.9376 | 10.0576 | 9 9176 | 10.0776 | V |
| Y5-Y8 | Absolute error | XXV | X | -9.9727 | -10 0127 | -9.9727 | -10.0127 | -9.9327 | -10.0527 | -9.9127 | -10 0722 | V |
| Z1-Z3 | Power supply filter | XXVI | S | 0.08 | 1.50 | 0.075 | 1.500 | 0.075 | 2.000 | 0.070 | 2 000 | µF |

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.





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4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Z in table I shall be performed in the order specified in 4 4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions G through L and N through Z shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4 Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition Z, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests G through L and N through Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests G through Z shall be performed in the order specified in 4 4.2 using the 25° ±5 C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6. Cross reference of classification. Part number M28787/232-1 supersedes part number M28787/232. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

| | |
|---|---|
| Maximum power dissipation | 2 W (outputs not loaded) |
| Supply voltages | +15 Vdc ± 5 percent, 50 mA maximum |
| | -15 Vdc ± 5 percent, 50 mA maximum |
| | +5 Vdc ± 10 percent, 75 mA maximum |
| Absolute maximum voltage supply ratings | ±16.5 Vdc, + 7 Vdc |
| Absolute maximum analog input voltage rating | ±25 V |
| Analog input impedance | 1[illegible] minimum |



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| | |
|---|---|
| Power supply noise rejection (referenced to input) | -30 dB minimum |
| Digital input voltage | |
| Low level | -0.5 V minimum, 0.7 V maximum |
| High level | 2.0 V minimum, 5.5 V maximum |
| Resolution | 1 LSB = 4.8828125 mV |
| Quantizing error | ±1/2 LSB |
| Hysteresis | |
| $V_{in}$ = 0 V | ±1 LSB maximum |
| $V_{in}$ ≠ 0 V | ±1/2 LSB maximum |
| Zero error | ±1 LSB maximum |
| Differential nonlinearity | ±1 LSB maximum |
| Linearity error | ±1 LSB maximum |
| Absolute error (% full scale) | ±0.4 percent maximum |
| Digital output codes | |
| Parallel | 2's complement, inverted analog<br>Offset binary, inverted analog |
| Serial | Offset binary, inverted analog, nonreturn to zero |
| Conversion rate | 3 µs/bit maximum |
| Analog input voltage range which results in a linear input/output | ±10 V |
| Multiplexed input delay (setting time to ±1 LSB) | 3 µs maximum |
| Clock input transition time | |
| $t_{TLH}$ (0.7 to 2.0 V points) | 100 ns maximum |
| $t_{THL}$ (0.7 to 2.0 V points) | 100 ns maximum |
| Clock pulse width | |
| Low level | 200 ns maximum |
| High level | 200 ns maximum |
| Clock frequency | 10 kHz minimum, 330 kHz maximum |

6.5 Module function. This module contains a single successive approximation analog to digital converter.

6.5.1 Input to output relationship. The QQQ module is a 12-bit successive approximation analog to digital converter which has a linear input-output relationship for a bipolar analog input voltage range of ±10 V. The applied analog input voltage is converted to an output digital code which is available as.

a. Serial output Offset binary, inverted analog, nonreturn to zero.
b. Parallel output. Offset binary, inverted analog.
c Parallel output 2's complement, inverted analog.



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The start convert signal initiates a reset by being pulsed low for at least 200 ns prior to a low-to-high clock transition. Once start convert initiates a reset, the converter will remain in the reset state as long as start convert is held low. As soon as start convert is brought high, a conversion will occur on the next low-to-high clock transition provided that start convert was brought high at least 200 ns prior to the clock transition. The conversion will occur at a rate of one bit for each low-to-high clock transition, beginning with the most significant bit (MSB). A conversion can be stopped and the converter reset at any time by pulsing start convert low and meeting the above requirements. An end-of-convert (EOC) signal appears at pin 32 when the 12th bit, referred to as the least significant bit (LSB), has been converted. The end-of-convert signal is a high-to-low transition of the logic signal at pin 32. When the end-of-convert signal is in a high state, the module is in either a reset state or is in the process of a conversion. If the end-of-convert signal is in a low state, the parallel digital outputs are available from the previous conversion. Continuous conversion is possible by connecting the EOC signal (pin 32) to start convert (pin 33). Several return lines are available to allow connection of signal and power supply returns to an internal unipoint ground. The input to output relationship is defined in table II. All digital outputs must interface with logic which has level sensitive inputs.

6.5.2 Paralleling of outputs. Paralleling of outputs is not permissible.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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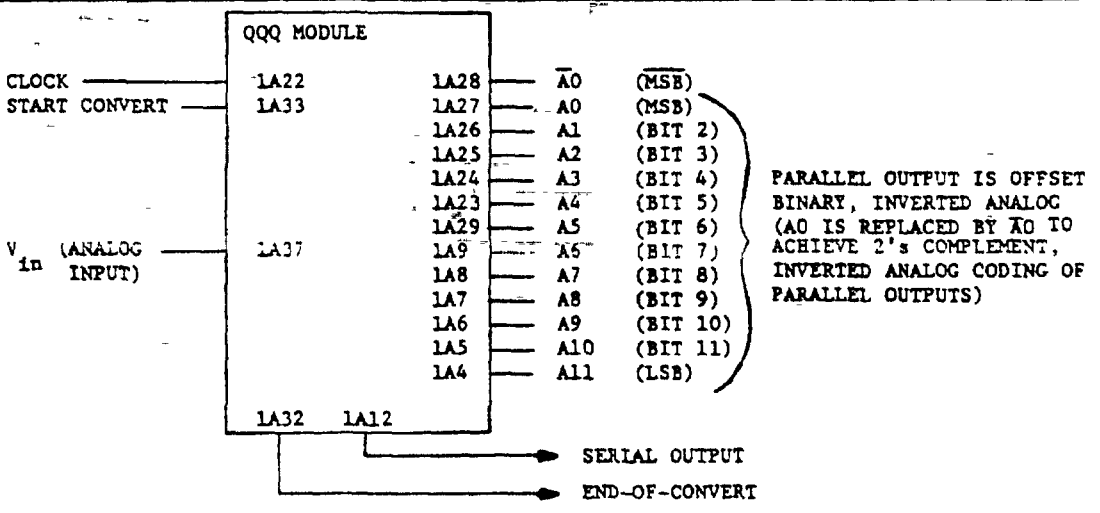
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TABLE II. Input to output relationship.

| Analog input voltage $V_{in}$ (V) | Digital counts in LSB's | Digital output code | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | $\overline{A0}$ | A0 | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 |
| +10.0 | 0 | H | L | L | L | L | L | L | L | L | L | L | L | L |
| +7.5 | 512 | H | L | L | H | L | L | L | L | L | L | L | L | L |
| +5.0 | 1024 | H | L | H | L | L | L | L | L | L | L | L | L | L |
| +2.5 | 1536 | H | L | H | H | L | L | L | L | L | L | L | L | L |
| 0.0 | 2048 | L | H | L | L | L | L | L | L | L | L | L | L | L |
| -2.5 | 2560 | L | H | L | H | L | L | L | L | L | L | L | L | L |
| -5.0 | 3072 | L | H | H | L | L | L | L | L | L | L | L | L | L |
| -7.5 | 3584 | L | H | H | H | L | L | L | L | L | L | L | L | L |
| -9.99512 | 4095 | L | H | H | H | H | H | H | H | H | H | H | H | H |




NOTES

1. $V_{in} = 10 \left[ 1 - \overline{A0} - \frac{A1}{2} - \frac{A2}{2^2} - \frac{A3}{2^3} - \frac{A4}{2^4} - \frac{A5}{2^5} - \frac{A6}{2^6} - \frac{A7}{2^7} - \frac{A8}{2^8} - \frac{A9}{2^9} - \frac{A10}{2^{10}} - \frac{A11}{2^{11}} \right]$ volts.

2. $V_{in} = \left[ 10 - (0.0048828125 \times \text{digital counts in LSB's}) \right]$ volts.

3. Digital counts in LSB's $= 2^{11}\,\overline{A0} + 2^{10}\,A1 + 2^{9}\,A2 + 2^{8}\,A3 + 2^{7}\,A4 + 2^{6}\,A5 + 2^{5}\,A6 + 2^{4}\,A7 + 2^{3}\,A8 + 2^{2}\,A9 + 2^{1}\,A10 + A11.$

4. A0 through A11 have the values 0 (low level) or 1 (high level) which corresponds to the low or high levels of each digital output

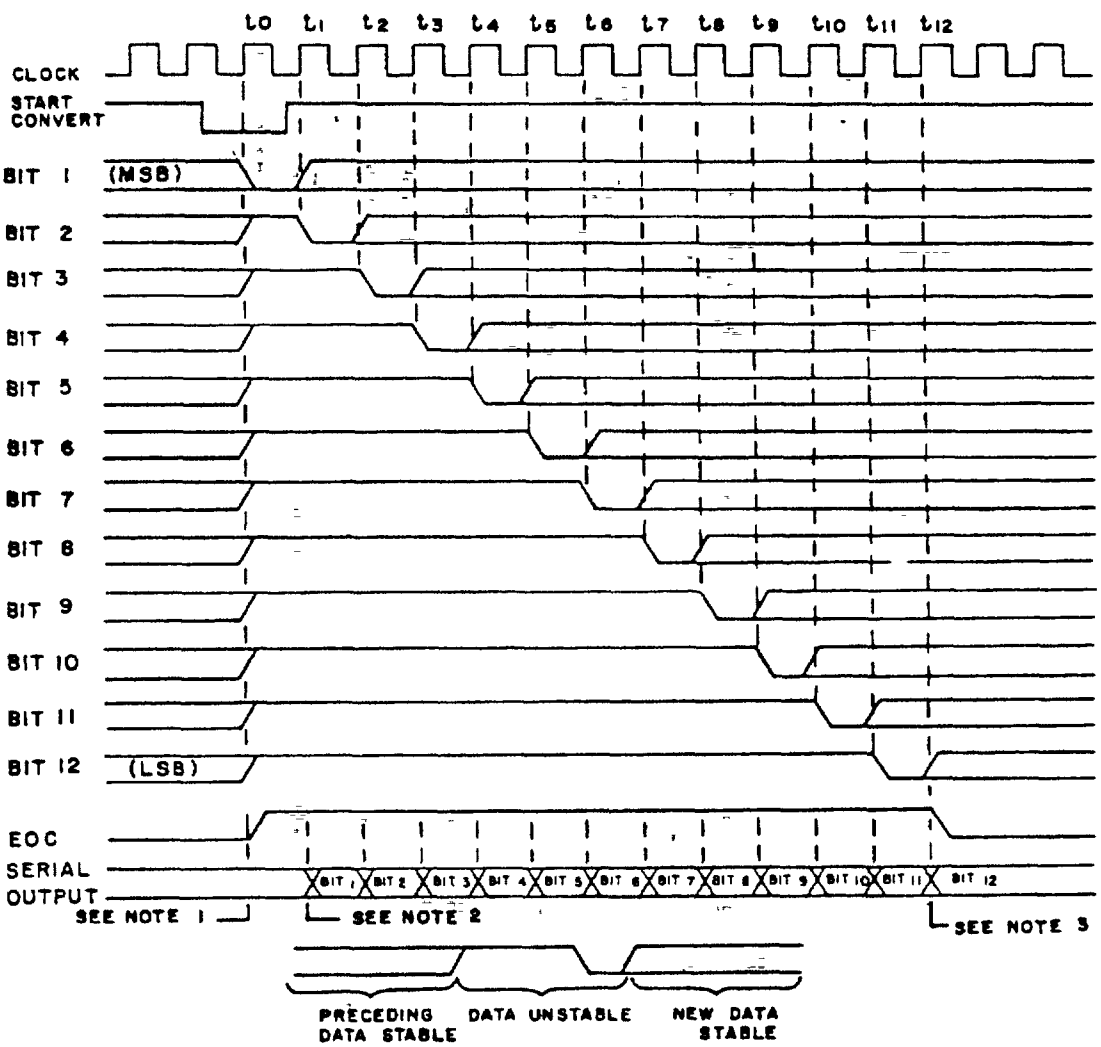
5. H = High-voltage level.

6. L = Low-voltage level.

7. Figure 2 shows the timing diagram for a single conversion.



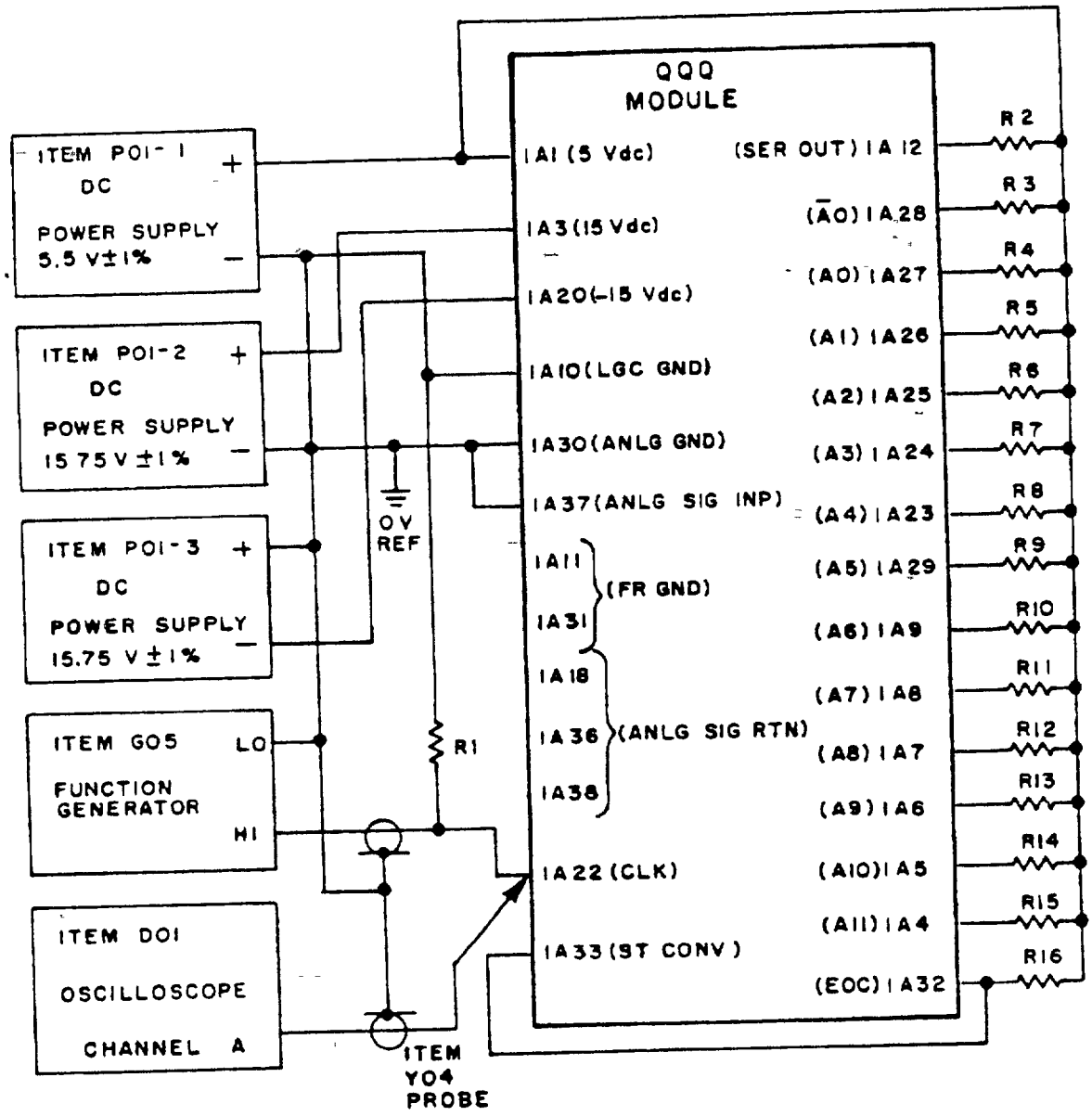
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:49Z
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NOTES:
1. Reset occurs.
2. Conversion begins after start convert is brought high and occurs on the next low-to-high clock transition.
3. Conversion complete

FIGURE 2 Timing diagram for a single conversion

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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 - R16 = 1.58 kΩ ±1%, 1/2 W.
3. Item GO5 setup:
   Square wave output.
   Frequency = 200 kHz ±25%.
   High level = 4.0 ±1.0 V.
   Low level = 0.0 ±0.2 V

FIGURE 3. Life, operating temperature, and component temperature test circuit



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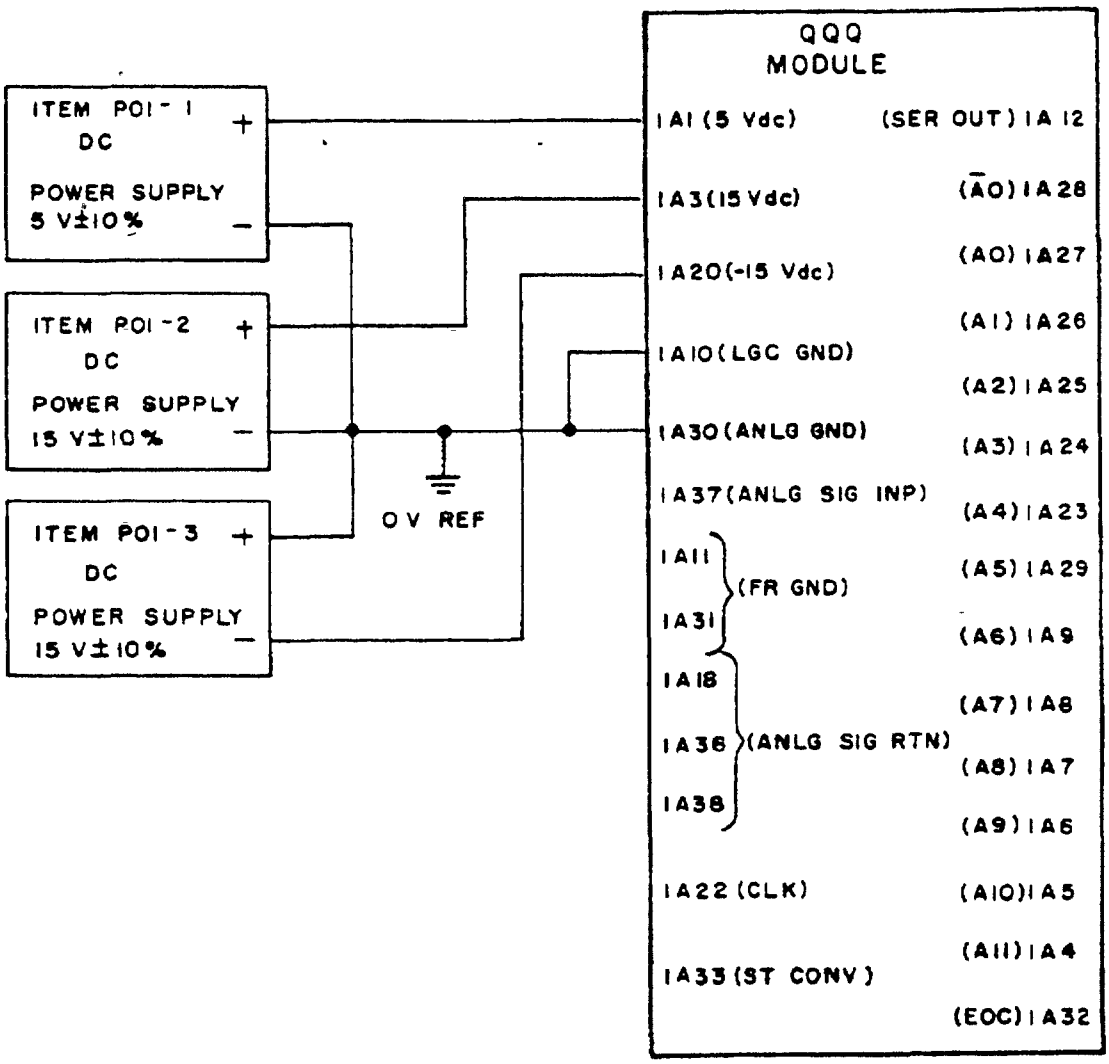



FIGURE 4. Vibration test circuit

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TABLE III. Isolation test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| A1 | Isolation, frame | 2,13,14,15,16, 17,19,21,34, 35,39,40,C | 11, 31 | 10.0 | - | MΩ | 5 |
| A2 | Isolation, pin | 2,13,14,15,19, 21,11,31,C | 16,17,34,35, 39,40 | | | | |
| A3 | | 2,14,17,19,21, 34,39 | 13,15,16,35, 40,11,31,C | | | | |
| A4 | | 13,14,16,17, 21,34,35 | 2,15,19,39,40, 11,31,C | | | | |
| A5 | | 2,16,17,21,11, 31,C | 14,15,19,34, 35,40 | | | | |





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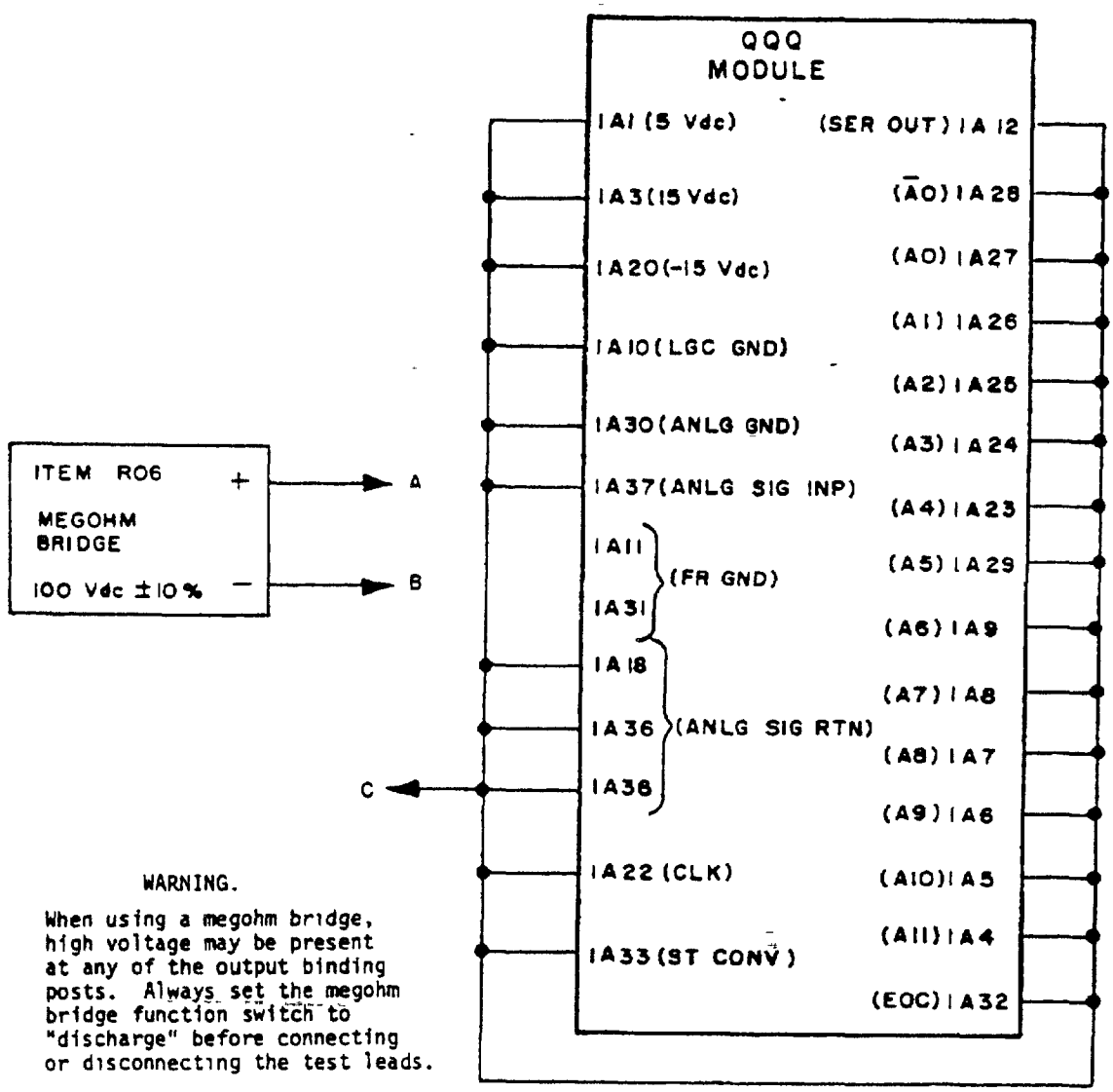



FIGURE 5 Isolation test circuit for table III

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TABLE IV. Continuity test.

| Test condition | Parameter | Connect | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| B1 | Continuity, analog return | Open | Open | 18 | 30 | 0.0 | 1.0 | Ω | 6 |
| B2 | Continuity, analog return | ↓ | ↓ | 36 | 30 | ↓ | ↓ | ↓ | ↓ |
| B3 | Continuity, analog return | | | 38 | 30 | | | | |
| B4 | Continuity, frame pin to frame pin | | | 11 | 31 | | | | |
| B5 | Continuity, fin to frame pin | | | 11, 31 | Module fin | | | | |
| B6 | Continuity, analog return | 18 | 30 | Open | Open | | 0.1 | | |
| B7 | Continuity, analog return | 36 | 30 | Open | Open | | 0.1 | | |
| B8 | Continuity, analog return | 38 | 30 | Open | Open | | 0.1 | | |





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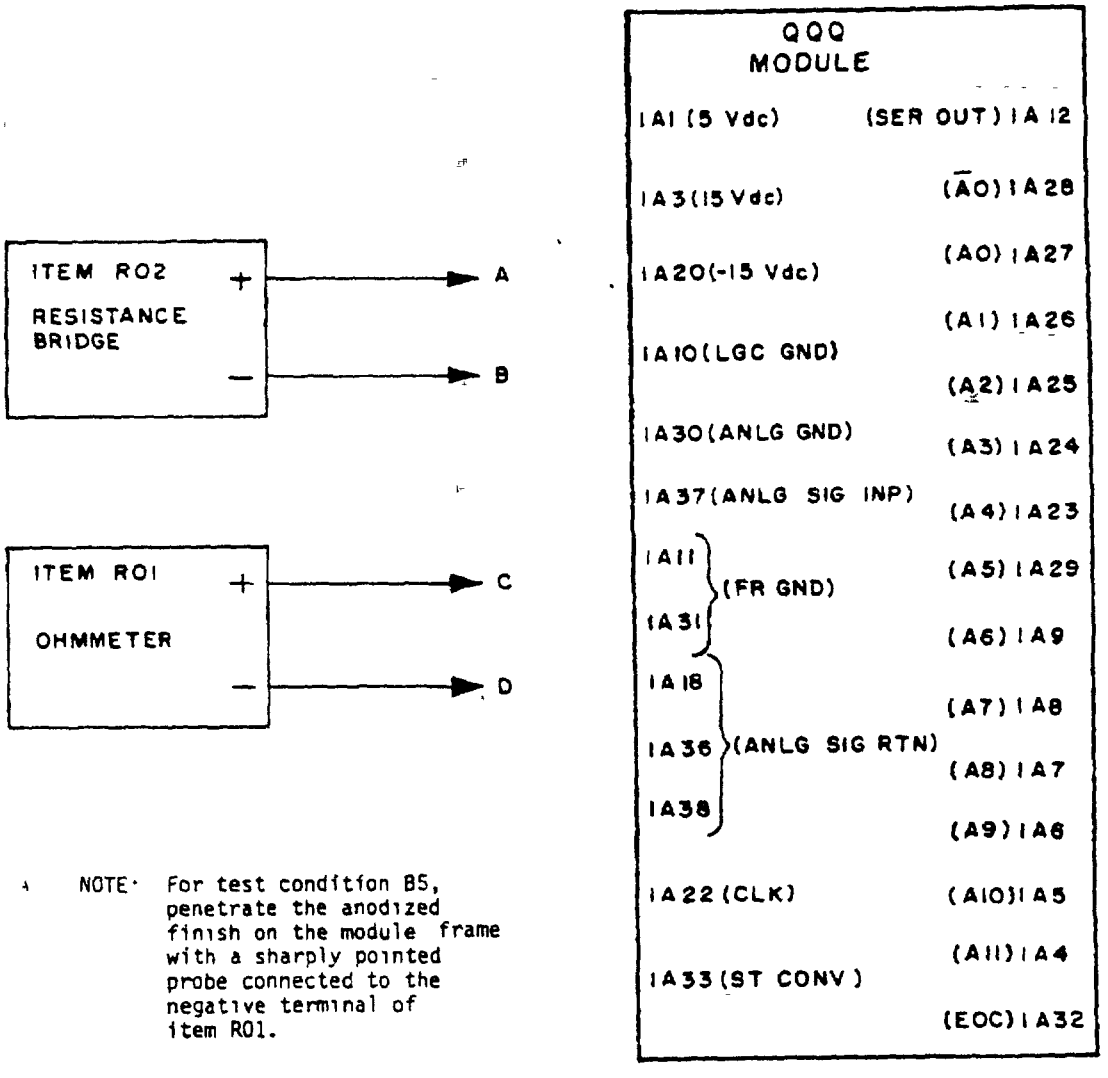
TABLE V. Isolation test.

| Test condition | Parameter | Connect | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| C | Isolation, digital ground to analog ground | Open | Open | 10 | 30 | 1 | - | kΩ | 6 |





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NOTE: For test condition B5, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of item RO1.

FIGURE 6. Continuity and isolation test circuit for tables IV and V.



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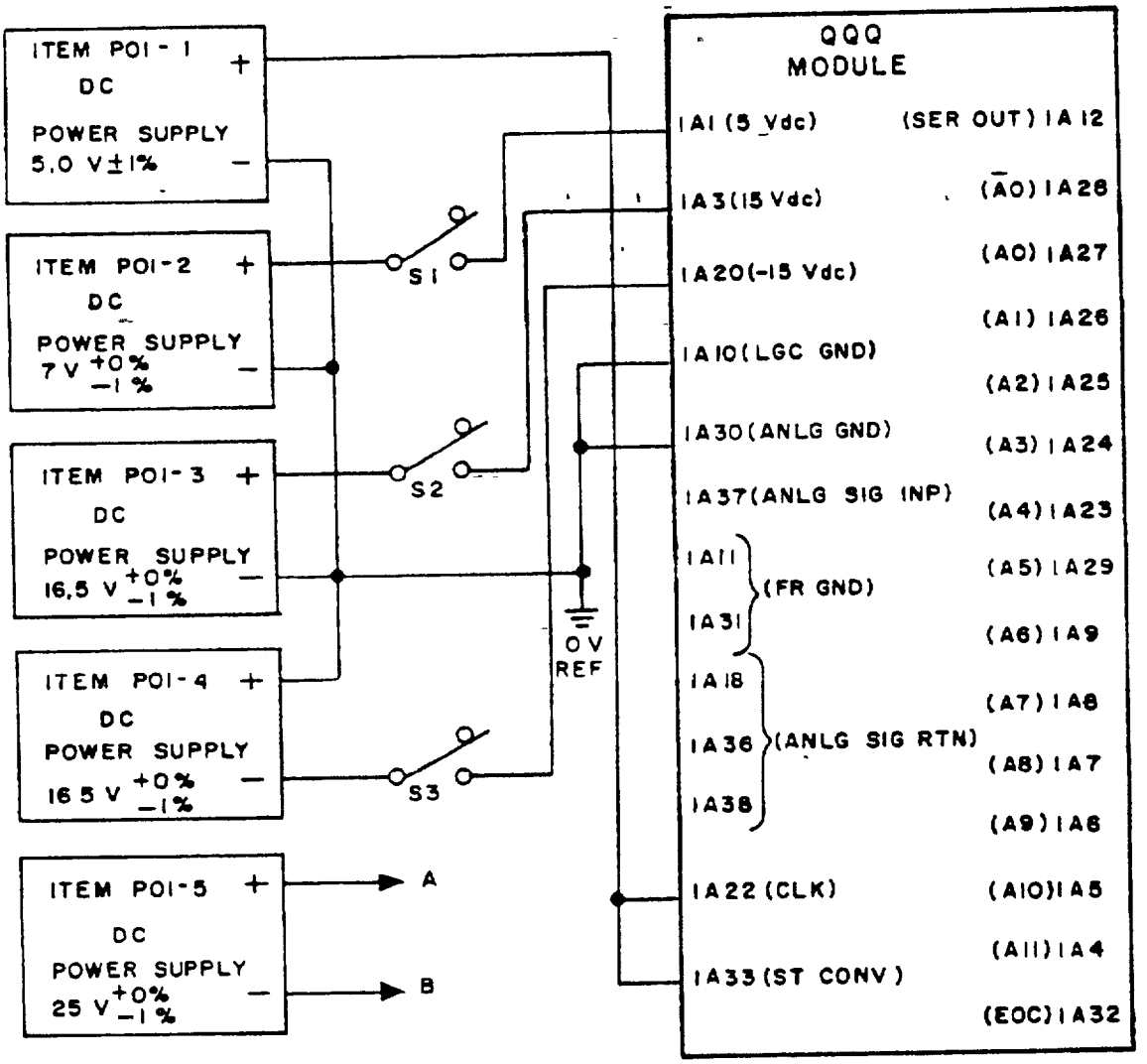
TABLE VI. Power supply sequencing test.

| Test condition | Parameter | Connect | | Switch positions | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | S1 | S2 | S3 | Min | Max | | |
| D1 | Power supply sequencing | Open | Open | Closed | Open | Open | 5 | 60 | s | 7 |
| D2 | | | | Open | Closed | Open | | | | |
| D3 | | | | Open | Open | Closed | | | | |
| D4 | | | | Open | Closed | Closed | | | | |
| D5 | | | | Closed | Open | Closed | | | | |
| D6 | | | | Closed | Closed | Open | | | | |
| D7 | | 37 | 30 | Closed | Closed | Closed | | | | |
| D8 | | 37 | 30 | Open | Open | Open | | | | |
| D9 | | 30 | 37 | Open | Open | Open | | | | |
| D10 | | 30 | 37 | Closed | Closed | Closed | | | | |





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NOTES
1. S1 = S2 = S3 = SPDT switch.
2. Switches S1-S3 shall remain in the position specified for the appropriate test condition for the times required by table VI. Module shall pass the remainder of this specification as proof of no damage.

FIGURE 7. Power supply sequencing test circuit for table VI.



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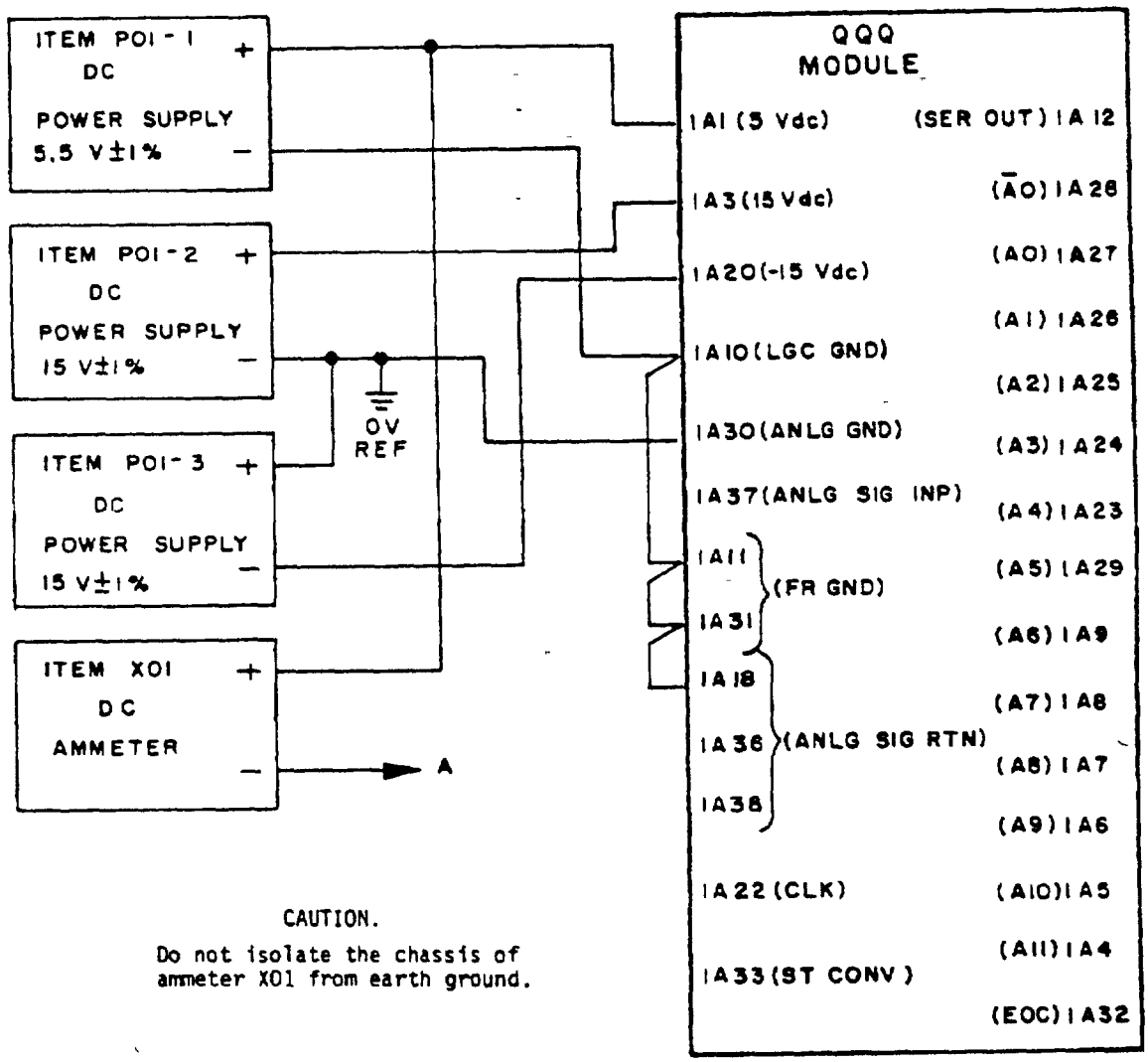
TABLE VII. Input breakdown current test.

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| E1 | Input breakdown current | 22 | 0 | 1 | mA | 8 |
| E2 | Input breakdown current | 33 | 0 | 1 | mA | 8 |





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CAUTION.

Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 8. Input breakdown current test circuit for table VII.



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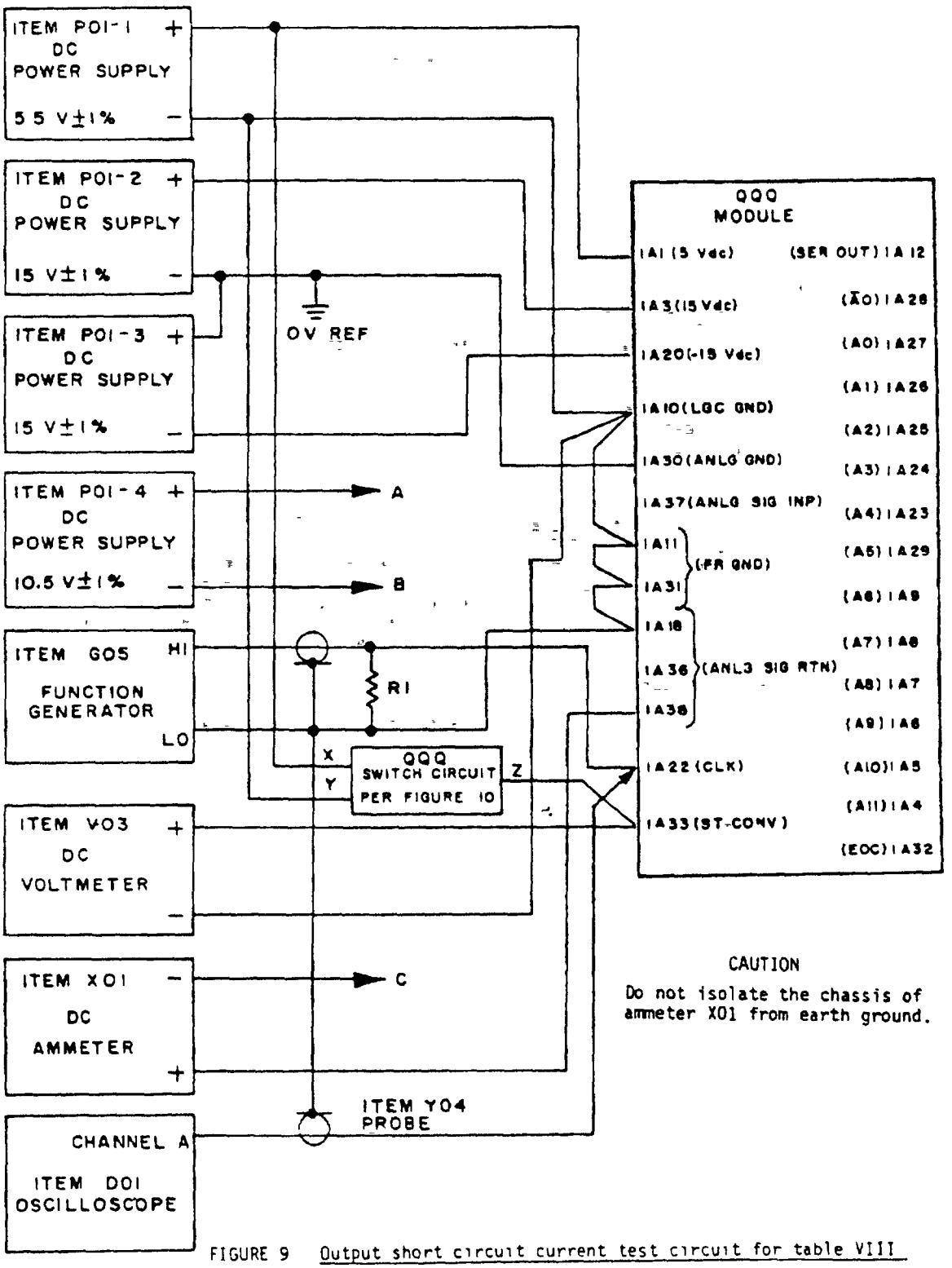
TABLE VIII. Output short circuit current test.

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| F1 | Output short circuit current | 37 | 36 | 28 | 0 | -25 | mA | 9 |
| F2 | | 36 | 37 | 27 | | | | |
| F3 | | | | 26 | | | | |
| F4 | | | | 25 | | | | |
| F5 | | | | 24 | | | | |
| F6 | | | | 23 | | | | |
| F7 | | | | 29 | | | | |
| F8 | | | | 9 | | | | |
| F9 | | | | 8 | | | | |
| F10 | | | | 7 | | | | |
| F11 | | | | 6 | | | | |
| F12 | | | | 5 | | | | |
| F13 | | | | 4 | | | | |
| F14 | | | | 12 | | | | |
| F15 | | | | 32 | | | | |





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CAUTION

Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 9 Output short circuit current test circuit for table VIII



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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. When performing the output short circuit test only one output shall be shorted at a time. The duration of each test shall not exceed 5 seconds.
3. Item G05 setup:
   Square wave output.
   Frequency = 200 kHz ±25%
   High level = 4.0 ±1.0 V.
   Low level = 0.0 ±0.2 V.
4. Test procedure for test condition F1.

   a. Connect the test circuit equipment as specified (do not connect point C at this time) for the appropriate test condition.
   b. Set switch S1 of figure 10 to position 2 and allow at least one low-to-high clock pulse at pin 22. The voltage reading on item V03 shall be 0.0 to 0.4 V.
   c. Set switch S1 of figure 10 to position 1 and allow at least 12 low-to-high clock pulses at pin 22. The voltage reading on item V03 shall be 2 4 to 5.5 V.
   d. Connect point C as specified in table VIII. The current measured on ammeter X01 shall satisfy the limits of table VIII.

5. Test procedure for test conditions F2 - F14:

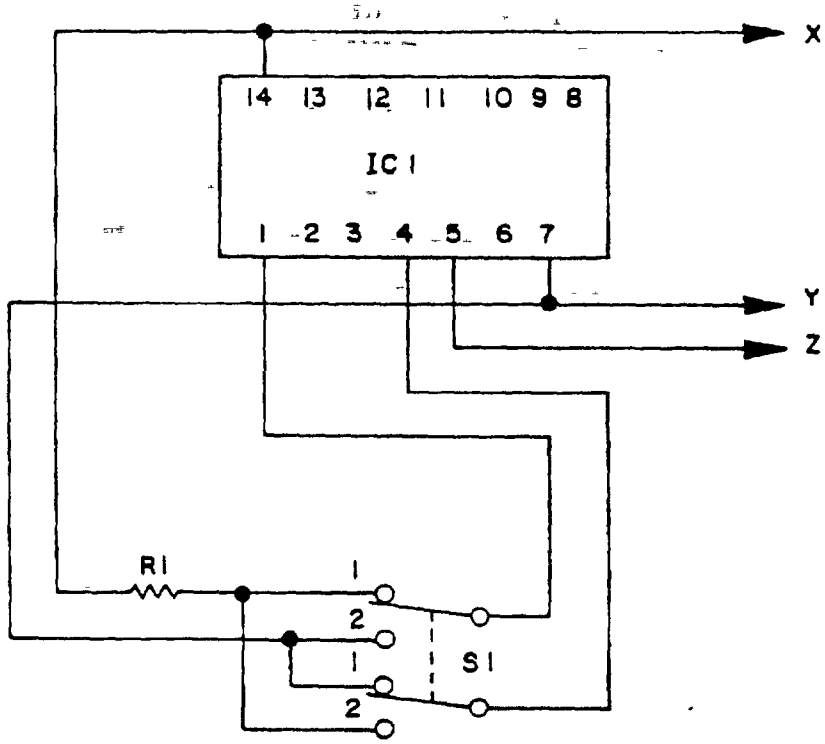
   a. Repeat steps 4a - 4d. Steps 4b and 4c are only required once for test conditions F2 - F14.

6. Test procedure for test condition F15.

   a. Repeat steps 4a, 4b, and 4d.

FIGURE 9. Output short circuit current test circuit for table VIII - Continued

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NOTES.
1. R1 = 1 kΩ ±5%, 1/4 W.
2. S1 = DPDT switch
3. IC1 = 7474, or equivalent.

FIGURE 10. QQQ switch circuit



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*TABLE IX. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test.

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| G1 | $\overline{A0}$ @ t0 | 28 | 5 | 100 | ns | 11,12 |
| G2 | A0 @ t1 | 27 | | 100 | | |
| G3 | A1 @ t2 | 26 | | 60 | | |
| G4 | A2 @ t3 | 25 | | | | |
| G5 | A3 @ t4 | 24 | | | | |
| G6 | A4 @ t5 | 23 | | | | |
| G7 | A5 @ t6 | 29 | | | | |
| G8 | A6 @ t7 | 9 | | | | |
| G9 | A7 @ t8 | 8 | | | | |
| G10 | A8 @ t9 | 7 | | | | |
| G11 | A9 @ t10 | 6 | | | | |
| G12 | A10 @ t11 | 5 | | | | |
| G13 | A11 @ t12 | 4 | | | | |
| G14 | EOC @ t0 | 32 | | | | |
| G15 | SER OUT @ t2 | 12 | | | | |

Transition time ($t_{TLH}$), $C_L$ = 50 pF





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*TABLE IX. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| G16 | $\overline{A0}$ @ t0 | 28 | 10 | 200 | ns | 11, 12 |
| G17 | A0 @ t1 | 27 | | 200 | | |
| G18 | A1 @ t2 | 26 | | 100 | | |
| G19 | A2 @ t3 | 25 | | | | |
| G20 | A3 @ t4 | 24 | | | | |
| G21 | A4 @ t5 | 23 | | | | |
| G22 | A5 @ t6 | 22 | | | | |
| G23 | A6 @ t7 | 9 | | | | |
| G24 | A7 @ t8 | 8 | | | | |
| G25 | A8 @ t9 | 7 | | | | |
| G26 | A9 @ t10 | 6 | | | | |
| G27 | A10 @ t11 | 5 | | | | |
| G28 | A11 @ t12 | 4 | | | | |
| G29 | EOC @ t0 | 32 | | | | |
| G30 | SER OUT @ t2 | 12 | | | | |

Transition time ($t_{TLH}$), $C_L$ = 150 pF





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*TABLE IX Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| G31 | $\overline{A0}$ @ t1 | 28 | 2 | 50 | ns | 11,12 |
| G32 | A0 @ t0 | 27 | | 50 | | |
| G33 | A1 @ t1 | 26 | | 30 | | |
| G34 | A2 @ t2 | 25 | | | | |
| G35 | A3 @ t3 | 24 | | | | |
| G36 | A4 @ t4 | 23 | | | | |
| G37 | A5 @ t5 | 29 | | | | |
| G38 | A6 @ t6 | 9 | | | | |
| G39 | A7 @ t7 | 8 | | | | |
| G40 | A8 @ t8 | 7 | | | | |
| G41 | A9 @ t9 | 6 | | | | |
| G42 | A10 @ t10 | 5 | | | | |
| G43 | A11 @ t11 | 4 | | | | |
| G44 | EOC @ t12 | 32 | | | | |
| G45 | SER OUT @ t2 | 12 | | | | |

Transition time ($t_{THL}$), $C_L$ = 50 pF





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*TABLE IX. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued

| Test condition | Parameter | | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | A t pin | Min | Max | | |
| G46 | $\overline{A0}$ @ t1 | Transition time ($t_{THL}$), $C_L$ = 150 pF | 2[illegible] | 5 | 100 | ns | 11,12 |
| G47 | A0 @ t0 | | 2[illegible] | | 100 | | |
| G48 | A1 @ t1 | | 26 | | 60 | | |
| G49 | A2 @ t2 | | 25 | | | | |
| G50 | A3 @ t3 | | 24 | | | | |
| G51 | A4 @ t4 | | 23 | | | | |
| G52 | A5 @ t5 | | 29 | | | | |
| G53 | A6 @ t6 | | 9 | | | | |
| G54 | A7 @ t7 | | 8 | | | | |
| G55 | A8 @ t8 | | 7 | | | | |
| G56 | A9 @ t9 | | 6 | | | | |
| G57 | A10 @ t10 | | 5 | | | | |
| G58 | A11 @ t11 | | 4 | | | | |
| G59 | EOC @ t12 | | 32 | | | | |
| G60 | SER OUT @ t2 | | 12 | | | | |





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*TABLE IX. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued

| Test condition | Parameter | | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | A to pin | Min | Max | | |
| C61 | CLK to $\bar{A}0$ @ t0 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 28 | 20 | 200 | ns | 11,12 |
| C62 | CLK to A0 @ t1 | | 27 | | 200 | | |
| C63 | CLK to A1 @ t2 | | 26 | | 160 | | |
| C64 | CLK to A2 @ t3 | | 25 | | | | |
| C65 | CLK to A3 @ t4 | | 24 | | | | |
| C66 | CLK to A4 @ t5 | | 23 | | | | |
| C67 | CLK to A5 @ t6 | | 29 | | | | |
| C68 | CLK to A6 @ t7 | | 9 | | | | |
| C69 | CLK to A7 @ t8 | | 8 | | | | |
| C70 | CLK to A8 @ t9 | | 7 | | | | |
| C71 | CLK to A9 @ t10 | | 6 | | | | |
| C72 | CLK to A10 @ t11 | | 5 | | | | |
| C73 | CLK to A11 @ t12 | | 4 | | | | |
| C74 | CLK to EOC @ t0 | | 32 | | | | |
| C75 | CLK TO SER OUT @ t2 | | 12 | | | | |





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*TABLE IX. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued

| Test condition | Parameter | | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | A to pin | Min | Max | | |
| G76 | CLK to A1 @ t0 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 26 | 20 | 160 | ns | 11,12 |
| G77 | CLK to A2 @ t0 | | 25 | | | | |
| G78 | CLK to A3 @ t0 | | 24 | | | | |
| G79 | CLK to A4 @ t0 | | 23 | | | | |
| G80 | CLK to A5 @ t0 | | 29 | | | | |
| G81 | CLK to A6 @ t0 | | 9 | | | | |
| G82 | CLK to A7 @ t0 | | 8 | | | | |
| G83 | CLK to A8 @ t0 | | 7 | | | | |
| G84 | CLK to A9 @ t0 | | 6 | | | | |
| G85 | CLK to A10 @ t0 | | 5 | | | | |
| G86 | CLK to A11 @ t0 | | 4 | | | | |
| G87 | CLK to $\overline{A0}$ @ t0 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 28 | | 220 | | |
| G88 | CLK to A0 @ t1 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 27 | | 220 | | |





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*TABLE IX. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| G89 | CLK to A1 @ t2 Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 26 | 20 | 180 | ns | 11,12 |
| G90 | CLK to A2 @ t3 | 25 | | | | |
| G91 | CLK to A3 @ t4 | 24 | | | | |
| G92 | CLK to A4 @ t5 | 23 | | | | |
| G93 | CLK to A5 @ t6 | 29 | | | | |
| G94 | CLK to A6 @ t7 | 9 | | | | |
| G95 | CLK to A7 @ t8 | 8 | | | | |
| G96 | CLK to A8 @ t9 | 7 | | | | |
| G97 | CLK to A9 @ t10 | 6 | | | | |
| G98 | CLK to A10 @ t11 | 5 | | | | |
| G99 | CLK to A11 @ t12 | 4 | | | | |
| G100 | CLK to EOC @ t0 | 32 | | | | |
| G101 | CLK to SER OUT @ t2 | 12 | | | | |
| G102 | CLK to A1 @ t0 | 26 | | | | |
| G103 | CLK to A2 @ t0 | 25 | | | | |





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*TABLE IX Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | A to pin | Min | Max | | |
| G104 | CLK to A3 @ t0 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 24 | 20 | 180 | ns | 11,12 |
| G105 | CLK to A4 @ t0 | | 23 | | | | |
| G106 | CLK to A5 @ t0 | | 29 | | | | |
| G107 | CLK to A6 @ t0 | | 9 | | | | |
| G108 | CLK to A7 @ t0 | | 8 | | | | |
| G109 | CLK to A8 @ t0 | | 7 | | | | |
| G110 | CLK to A9 @ t0 | | 6 | | | | |
| G111 | CLK to A10 @ t0 | | 5 | | | | |
| G112 | CLK to A11 @ t0 | | 4 | | | | |
| G113 | CLK to $\bar{A}0$ @ t1 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 28 | | 200 | | |
| G114 | CLK to A0 @t0 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 27 | | 200 | | |





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*TABLE IX. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| G115 | CLK to A1 @ t1 Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 26 | 20 | 160 | ns | 11,12 |
| G116 | CLK to A2 @t2 | 25 | | | | |
| G117 | CLK to A3 @ t3 | 24 | | | | |
| G118 | CLK to A4 @ t4 | 23 | | | | |
| G119 | CLK to A5 @ t5 | 29 | | | | |
| G120 | CLK to A6 @ t6 | 9 | | | | |
| G121 | CLK to A7 @ t7 | 8 | | | | |
| G122 | CLK to A8 @ t8 | 7 | | | | |
| G123 | CLK to A9 @ t9 | 6 | | | | |
| G124 | CLK to A10 @ t10 | 5 | | | | |
| G125 | CLK to A11 @ t11 | 4 | | | | |
| G126 | CLK to EOC @ t12 | 32 | | | | |
| G127 | CLK to SER OUT @ t2 | 12 | | | | |





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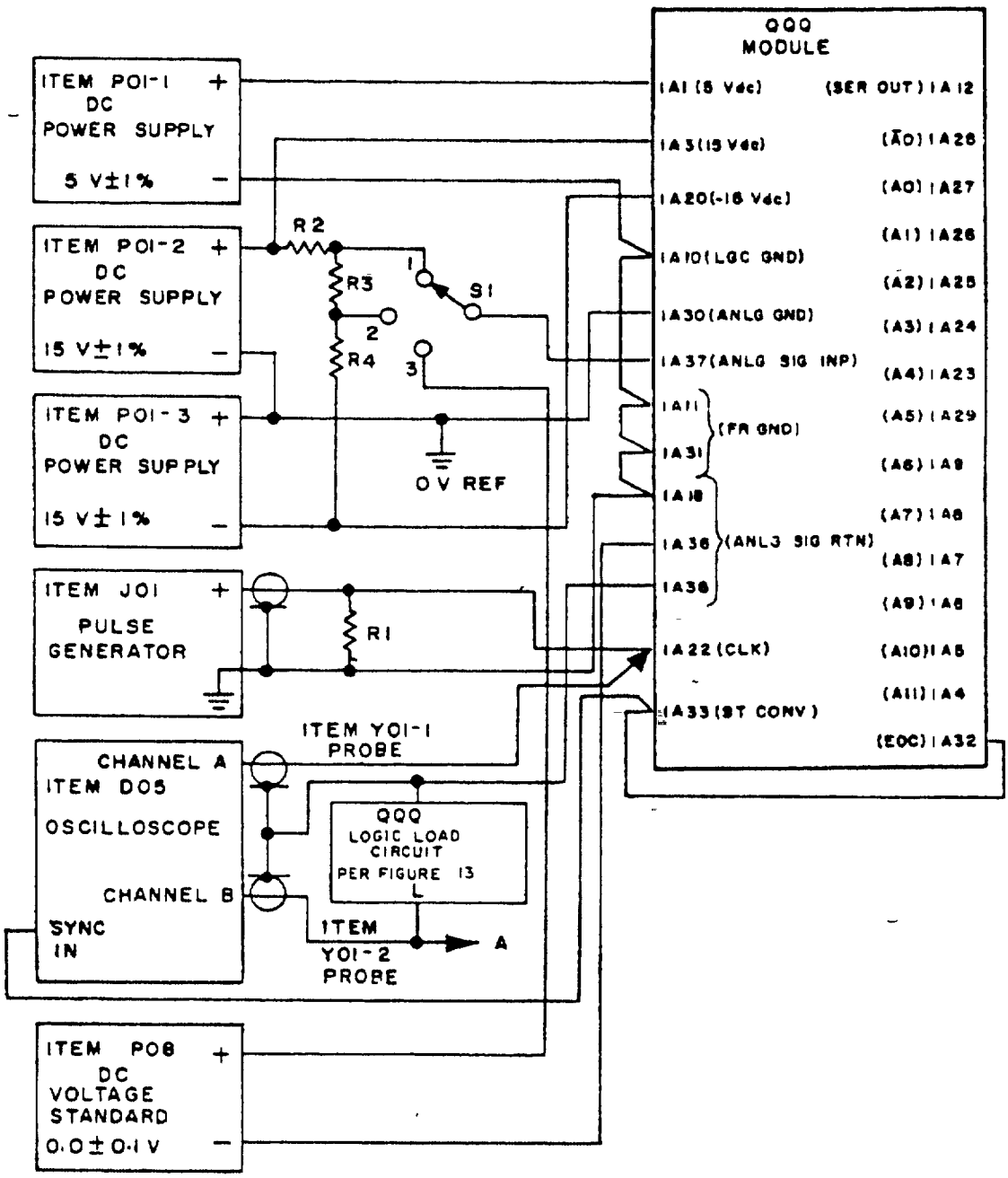
*TABLE IX. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| G128 | CLK to $\overline{A0}$ @ t1 Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | 28 | 20 | 220 | ns | 11,12 |
| G129 | CLK to A0 @ t0 | 27 | | 220 | | |
| G130 | CLK to A1 @ t1 | 26 | | 180 | | |
| G131 | CLK to A2 @ t2 | 25 | | | | |
| G132 | CLK to A3 @ t3 | 24 | | | | |
| G133 | CLK to A4 @ t4 | 23 | | | | |
| G134 | CLK to A5 @ t5 | 29 | | | | |
| G135 | CLK to A6 @ t6 | 9 | | | | |
| G136 | CLK to A7 @ t7 | 8 | | | | |
| G137 | CLK to A8 @ t8 | 7 | | | | |
| G138 | CLK to A9 @ t9 | 6 | | | | |
| G139 | CLK to A10 @ t10 | 5 | | | | |
| G140 | CLK to A11 @ t11 | 4 | | | | |
| G141 | CLK to EOC @ t12 | 32 | | | | |
| G142 | CLK to SER OUT @ t2 | 12 | | | | |





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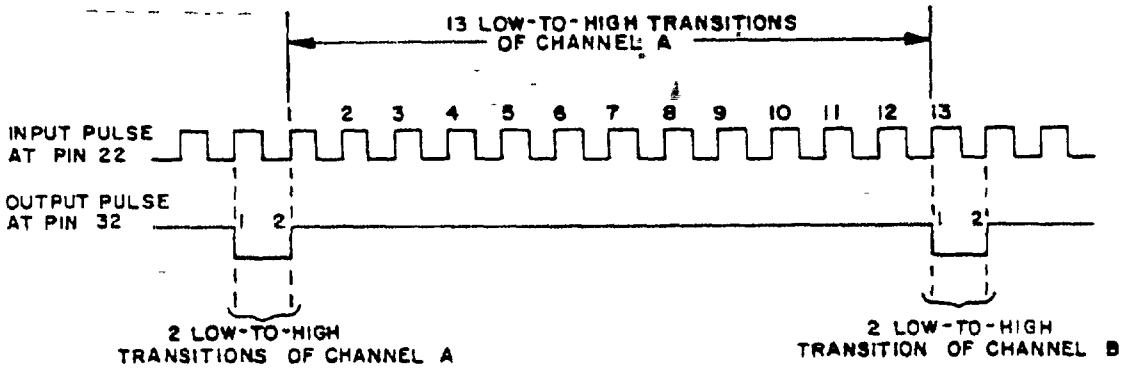



* FIGURE 11. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test circuit for table IX

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NOTES
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = R4 = 500Ω ±1%, 1/4 W.
3 R3 = 2.5 kΩ ±1%, 1/2 W.
4. S1 = SP 3 position switch.
5. Timing check (required prior to testing) per table IX

   a. Connect the equipment as specified for test condition G14.
   b. Set switch S1 to position 1. Adjust item D05 to obtain a display as shown below. If the display is not satisfied, the module shall be considered failed




6. Test procedure for test conditions G1-G14, G16-G29, G31-G44, G46-G59, G61-G74, G87-G100, G113-G126, and G128-G141

   a. Connect the equipment as specified for the appropriate test condition.
   b. Set switch S1 to position 2.
   c. Measurements shall satisfy the limits of table IX.

7. Test procedure for test conditions G76-G86 and G102-G112

   a Connect the equipment as specified for the appropriate test condition
   b Set switch S1 to position 1.
   c. Measurements shall satisfy the limits of table IX.

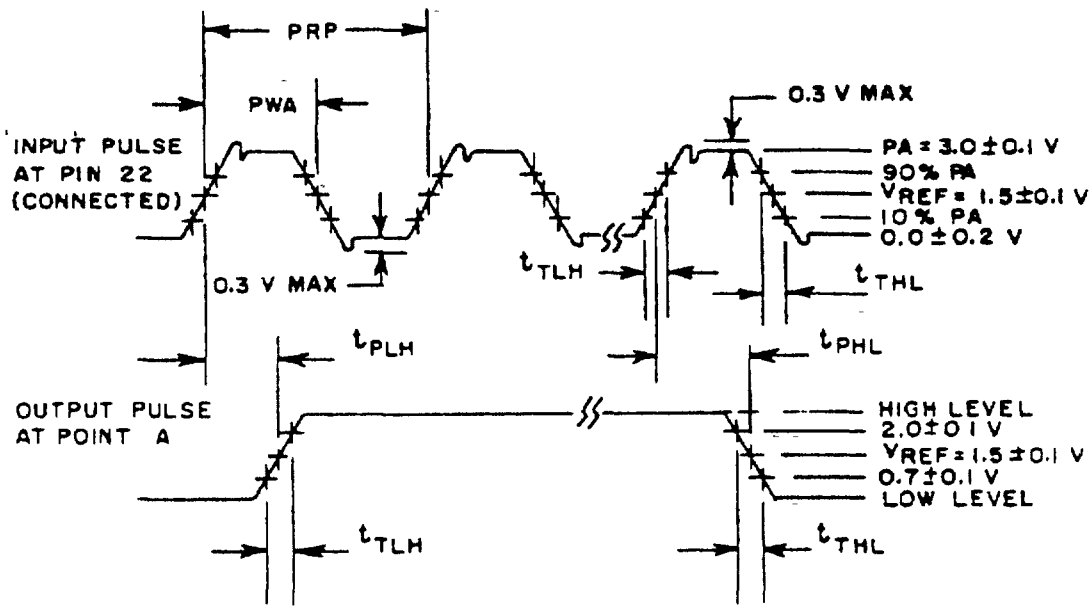
8. Test procedure for test conditions G15, G30, G45, G60, G75, G101, G127, and G142:

   a Connect the equipment as specified for the appropriate test condition.
   b. Set switch S1 to position 3.
   c. Vary item P08 setting until a stable pulse waveform oscilloscope display is observed.
   d. Measurements shall satisfy the limits of table IX.

FIGURE 11 Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test circuit for table IX - Continued



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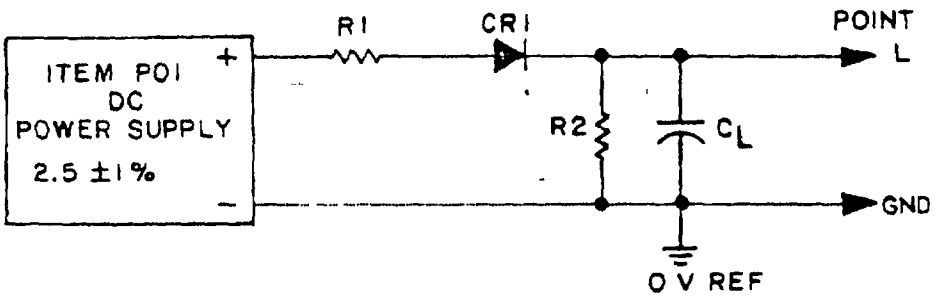



NOTES
1. Input pulse parameters
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PWA = 2.5 ±0.25 µs.
   PRP = 5.0 ±0.5 µs.
2. Before making the measurements specified in table IX, verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.3 V.
3. All voltages are referenced to 0 V unless otherwise specified.

* FIGURE 12. Transition time ($t_{TLH}$ and $t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$ test circuit waveforms for table IX



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NOTES
1. CR1 = 1N4150, or equivalent.
2 R1 = Select value for a current flow of 3.2 mA ±5% out of point L with point L held at 0.30 ±0.03 V.
3 R2 = 30.1 kΩ ±1%, 1/8 W.
4 $C_L$ = 50 ±5 pF (including probe and parasitic capacitance) for test conditions G1-G15, G31-G45, G61-G86, and G113-G127
5. $C_L$ = 150 ±5 pF (including probe and parasitic capacitance) for test conditions G16-G30, G46-G60, G87-G112, and G128-G142.
6. Capacitor $C_L$ is not required for test conditions K1-K30.
7 Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm)

FIGURE 13 QQQ logic load circuit.



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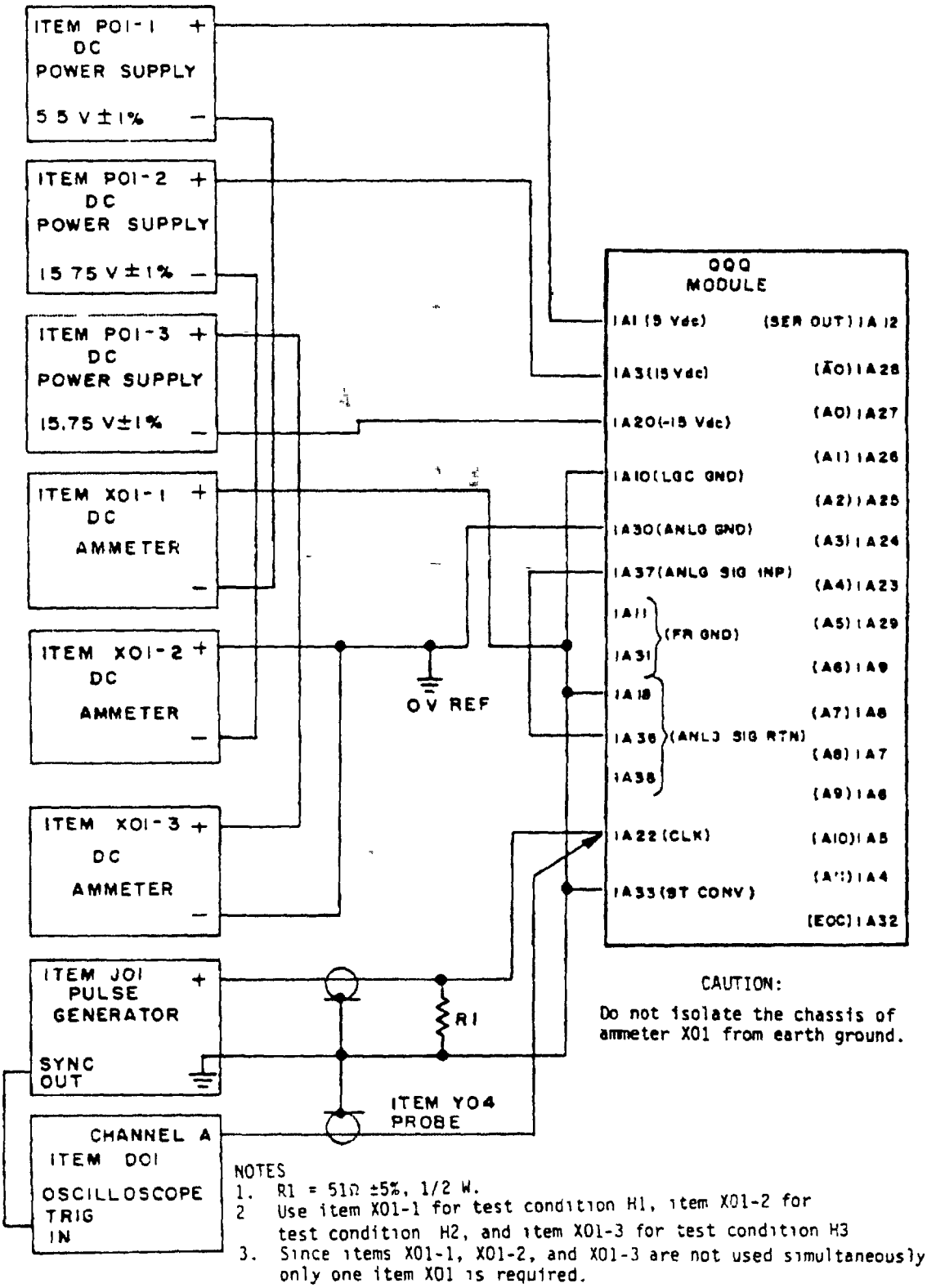
TABLE X. Power supply current test.

| Test condition | Parameter | Ammeter to be used | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | | Min | Max | | |
| H1 | Power supply current (5.5 Vdc) | XO1-1 | 1 | 75 | mA | 14, 15 |
| H2 | Power supply current (+15.75 Vdc) | XO1-2 | 1 | 50 | mA | 14, 15 |
| H3 | Power supply current (-15.75 Vdc) | XO1-3 | 1 | 50 | mA | 14, 15 |





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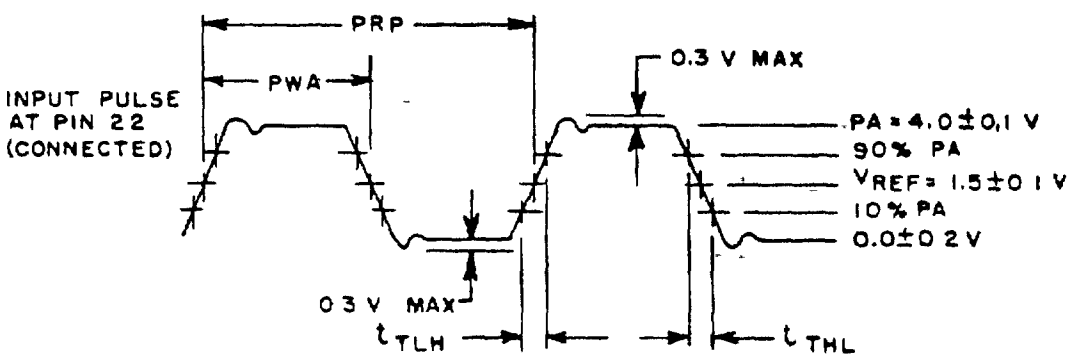



NOTES
1. R1 = 51Ω ±5%, 1/2 W.
2. Use item XO1-1 for test condition H1, item XO1-2 for test condition H2, and item XO1-3 for test condition H3
3. Since items XO1-1, XO1-2, and XO1-3 are not used simultaneously, only one item XO1 is required.

FIGURE 14 Power supply current test circuit for table X.



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NOTES
1. Input pulse parameters:
   $t_{TLH} = t_{THL}$ = 50 ±25 ns.
   PWA = 2.50 ±0.25 µs.
   PRP = 5.00 ±0.50 µs.
2. After making all circuit connections for a test condition, place item J01 in the single pulse mode. Apply two pulses to the circuit under test and then make the required measurement.

FIGURE 15 Power supply current test circuit waveforms for table X.



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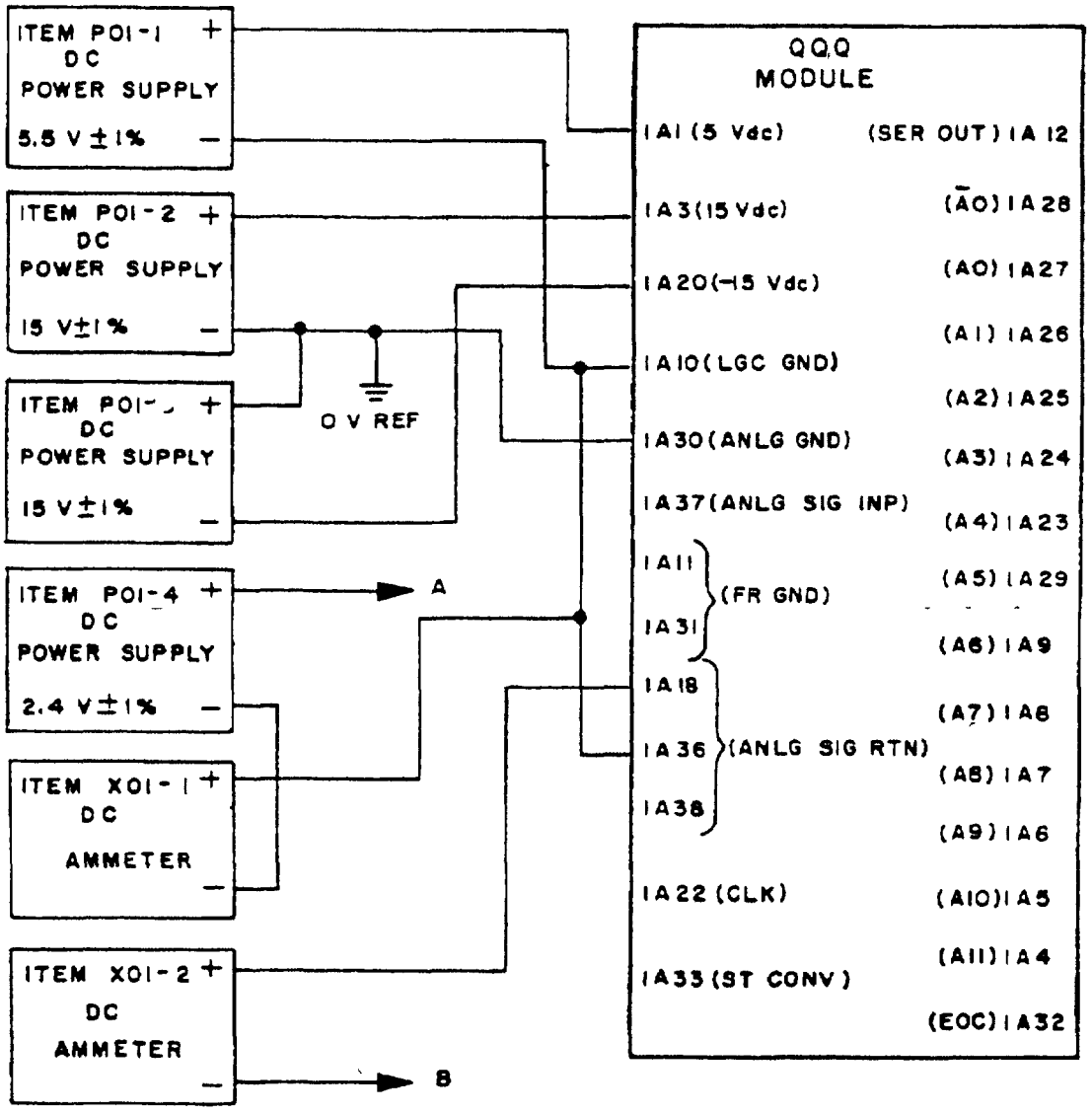
TABLE XI. Static input current (high-level and low-level) test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| J1 | Static input current, high level | 22 | Open | 0 | 20 | µA | 16 |
| J2 | Static input current, high level | 33 | Open | 0 | 20 | µA | |
| J3 | Static input current, low level | Open | 22 | -0.05 | -0.44 | mA | |
| J4 | Static input current, low level | Open | 33 | -0.05 | -0.44 | mA | |





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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.

FIGURE 16. Static input current (high-level and low-level) test circuit for table XI.

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TABLE XII. Static output voltage (high-level and low-level) test.

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| K1 | Static output voltage, high level | 37 | 36 | 28 | 2.4 | 4.5 | V | 17 |
| K2 | Static output voltage, low level | | | 27 | 0.0 | 0.3 | | |
| K3 | | | | 26 | | | | |
| K4 | | | | 25 | | | | |
| K5 | | | | 24 | | | | |
| K6 | | | | 23 | | | | |
| K7 | | | | 29 | | | | |
| K8 | | | | 9 | | | | |
| K9 | | | | 8 | | | | |
| K10 | | | | 7 | | | | |
| K11 | | | | 6 | | | | |
| K12 | | | | 5 | | | | |
| K13 | | | | 4 | | | | |
| K14 | | | | 12 | | | | |
| K15 | | | | 32 | | | | |
| K16 | | 36 | 37 | 28 | | | | |





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TABLE XII. Static output voltage (high-level and low-level) test - Continued.

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| K17 | Static output voltage, high level | 36 | 37 | 27 | 2.4 | 4.5 | V | 17 |
| K18 | | | | 26 | | | | |
| K19 | | | | 25 | | | | |
| K20 | | | | 24 | | | | |
| K21 | | | | 23 | | | | |
| K22 | | | | 29 | | | | |
| K23 | | | | 9 | | | | |
| K24 | | | | 8 | | | | |
| K25 | | | | 7 | | | | |
| K26 | | | | 6 | | | | |
| K27 | | | | 5 | | | | |
| K28 | | | | 4 | | | | |
| K29 | | | | 12 | | | | |
| K30 | | | | 32 | | | | |





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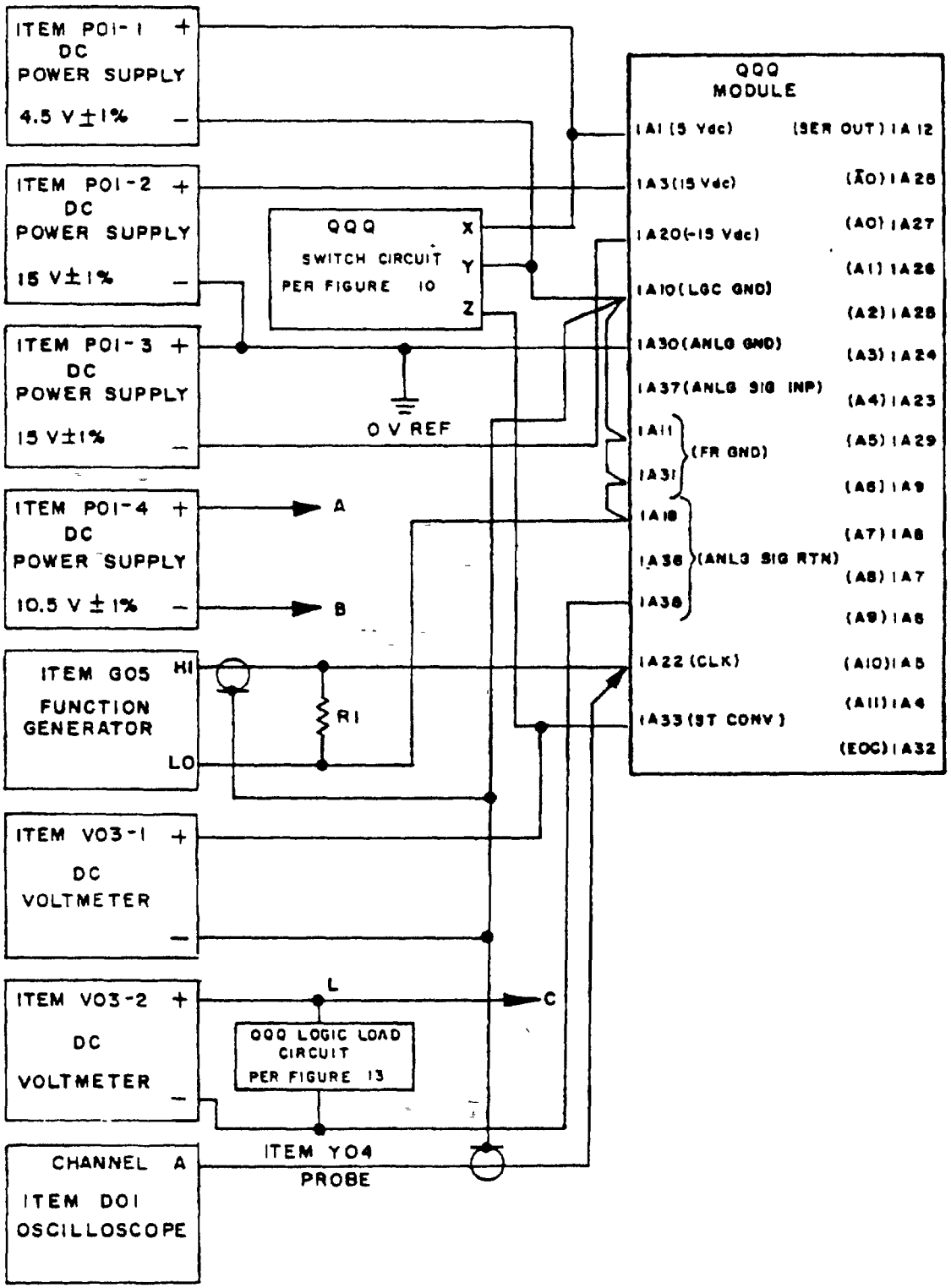



FIGURE 17 Static output voltage (high-level and low-level) test circuit for table XII.



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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Item G05 setup:
   Square wave output.
   Frequency = 200 kHz ±25%.
   High level = 4.0 ±1.0 V.
   Low level = 0.0 ±0.2 V.
3. Since items V03-1 and V03-2 are not used simultaneously, only one item V03 is required.
4. Test procedure for test conditions K1-K15

   a Connect the equipment as specified for the appropriate test condition.
   b Steps 4c and 4d are only required once for test conditions K1-K15.
   c. Set switch S1 of figure 10 to position 2 and allow at least one low-to-high clock pulse at pin 22 The voltage reading on item V03-1 shall be 0 0 to 0 4 V
   d Set switch S1 of figure 10 to position 1 and allow at least 12 low-to-high clock pulses at pin 22. The voltage reading on item V03-1 shall be 2.4 to 4.5 V.
   e. The voltage measured on item V03-2 shall satisfy the limits of table XII.

5. Test procedure for test conditions K16-K29: Repeat steps 4a - 4d.
6. Test procedure for test condition K30: Repeat steps 4a, 4c, and 4e.

FIGURE 17 Static output voltage (high-level and low-level) test circuit for table XII - Continued.



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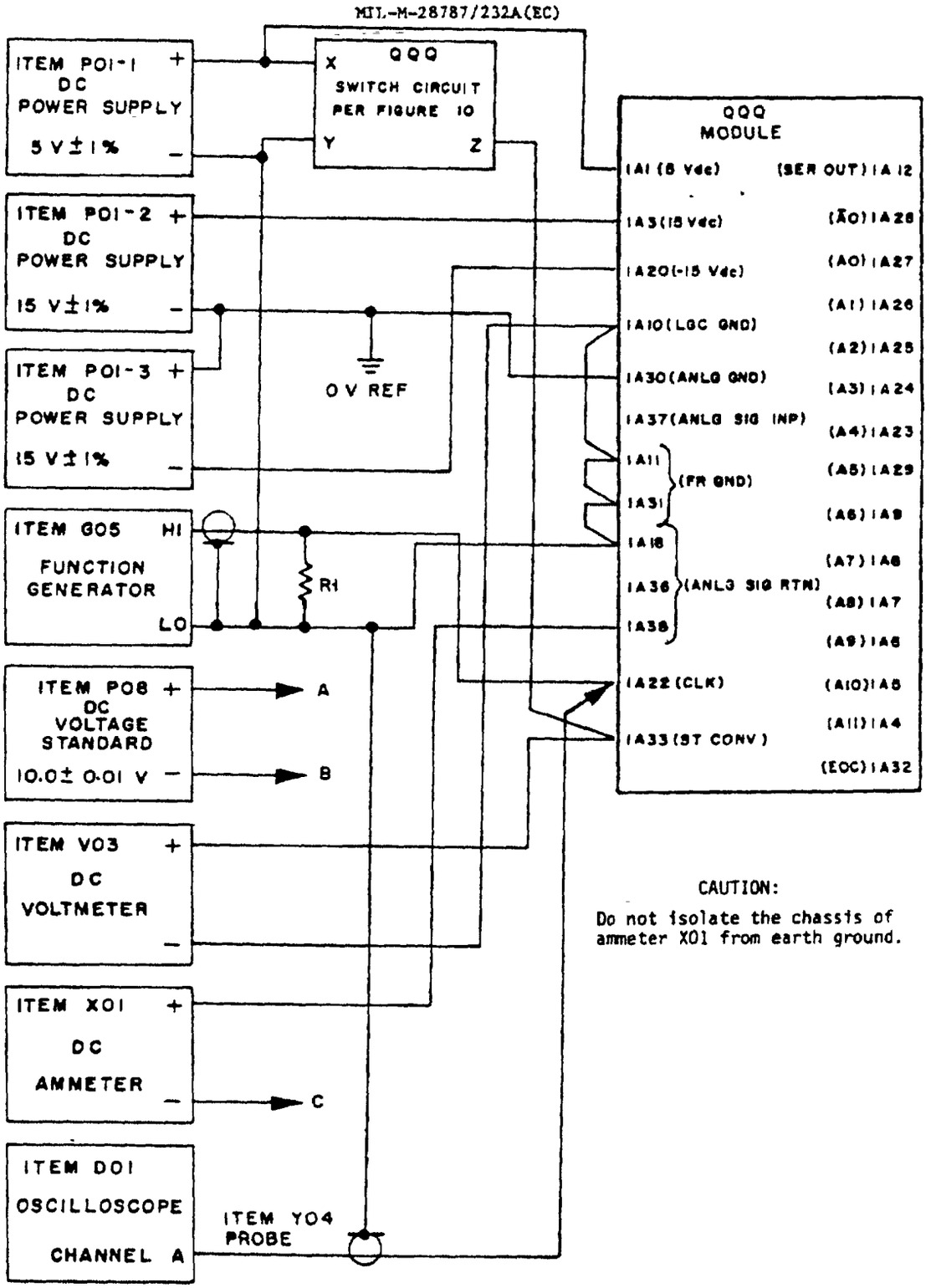
TABLE XIII. Input impedance check.

| Test condition | Parameter | Connect | | Initial limits 25° ±3°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| L1 | Input inpedance check (measuring input current) | 37 | C | 0 | 1 | mA | 18 |
| L2 | Input impedance check (measuring input current) | C | 37 | 0 | -1 | mA | 18 |





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CAUTION:
Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 18. Input impedance check test circuit for table VIII



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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Item G05 setup:
   Square wave output.
   Frequency = 200 kHz ±25%.
   High level = 4.0 ±1.0 V.
   Low level = 0.0 ±0.2 V.
3. Test procedure for test conditions L1 and L2.

   a. Connect the equipment as specified for the appropriate test condition.
   b. Set switch S1 of figure 10 to position 2 and allow at least one low-to-high clock pulse at pin 22. The voltage reading on item V03 shall be 0.0 to 0.4 V.
   c. Set switch S1 of figure 10 to position 1 and allow at least 12 low-to-high clock pulses at pin 22. The voltage reading on item V03 shall be 2.4 to 5.0 V.
   d. The current measured on item X01 shall satisfy the limits of table XIII.

FIGURE 18. Input impedance check test circuit for table XIII - Continued.



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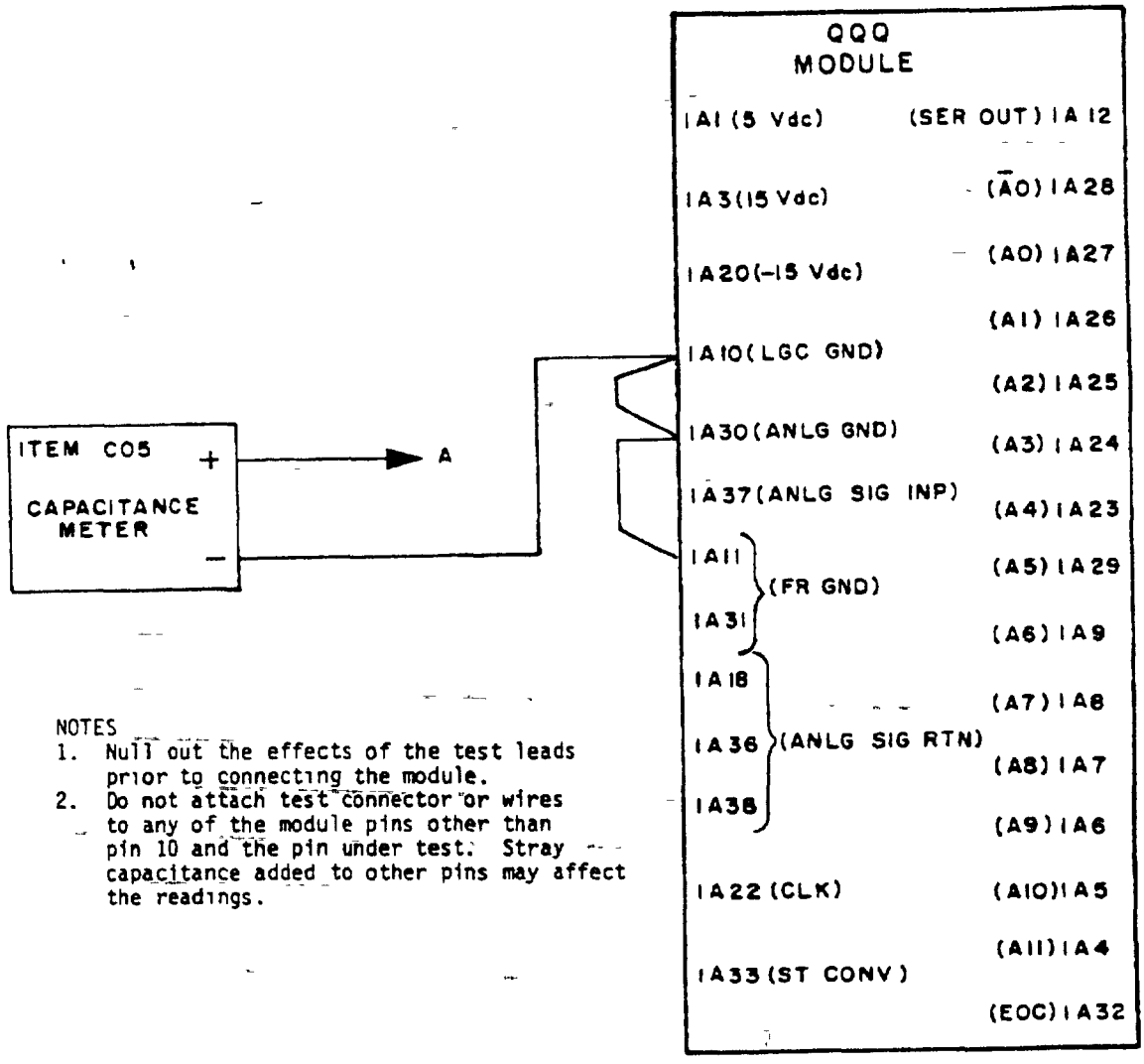
TABLE XIV. Input capacitance test.

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| M1 | Input capacitance | 22 | 2 | 15 | pF | 19 |
| M2 | Input capacitance | 33 | 2 | 15 | pF | 19 |





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NOTES
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 19. Input capacitance test circuit for table XIV.



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TABLE XV. Digital input threshold and pulse width check test.

| Test condition | Parameter | Connect A to pin | Connect B to pin | Digital output $\bar{A}0$ | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| N1 | Digital input threshold and pulse width check | 37 | 36 | H | L | L | L | L | L | L | L | L | L | L | L | Digital output | 20, 21 |
| N2 | | 36 | 37 | L | H | H | H | H | H | H | H | H | H | H | H | | |
| N3 | | 37 | 36 | H | L | L | L | L | L | L | L | L | L | L | L | | |
| N4 | | 36 | 37 | L | H | H | H | H | H | H | H | H | H | H | H | | |





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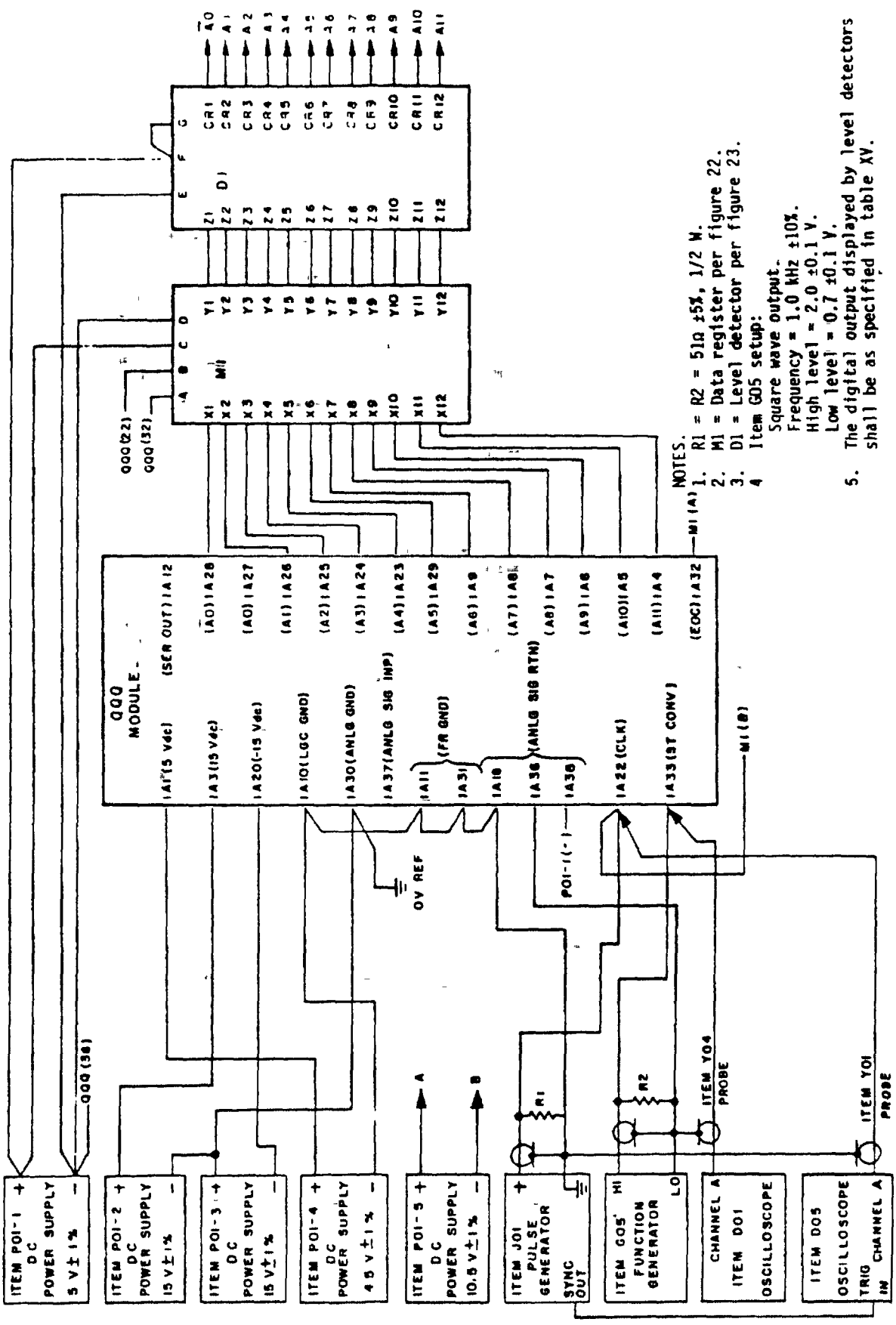
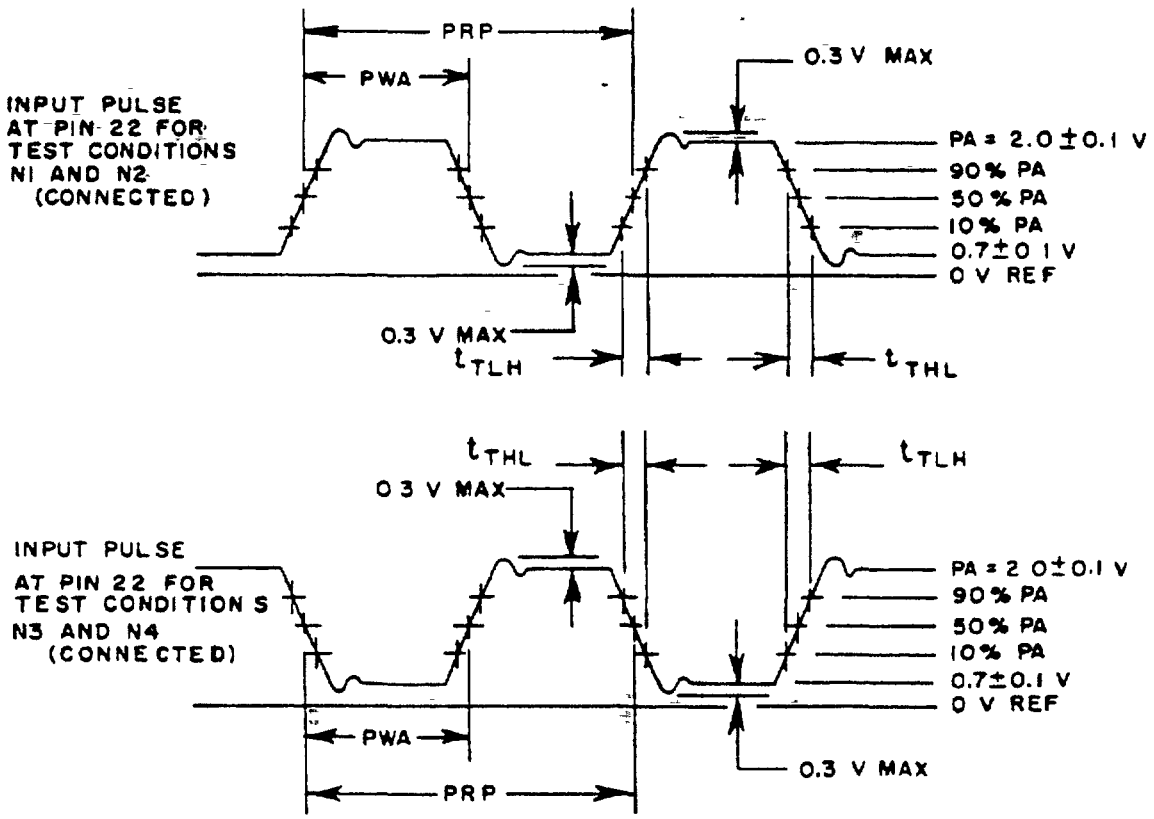



FIGURE 20. Digital input threshold and pulse width check test circuit for table XV.

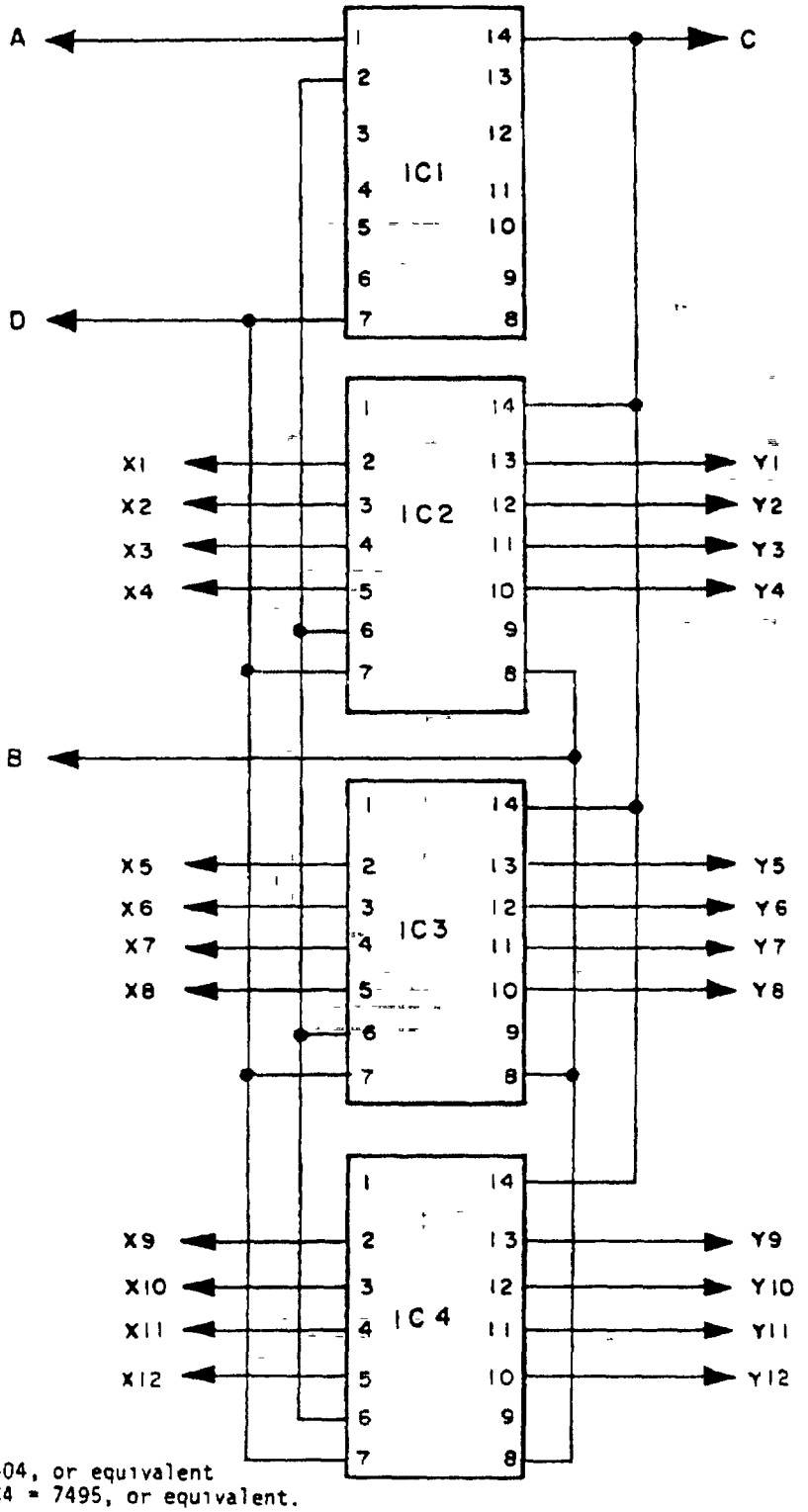
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Check the source to verify that this is the current version before use.




NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PWA = 200 ±10 ns.
   PRP = 5.0 ±0.5 µs.
2. All voltages are referenced to 0 V unless otherwise specified.

FIGURE 21 Digital input threshold and pulse width check test circuit waveforms for table XV.

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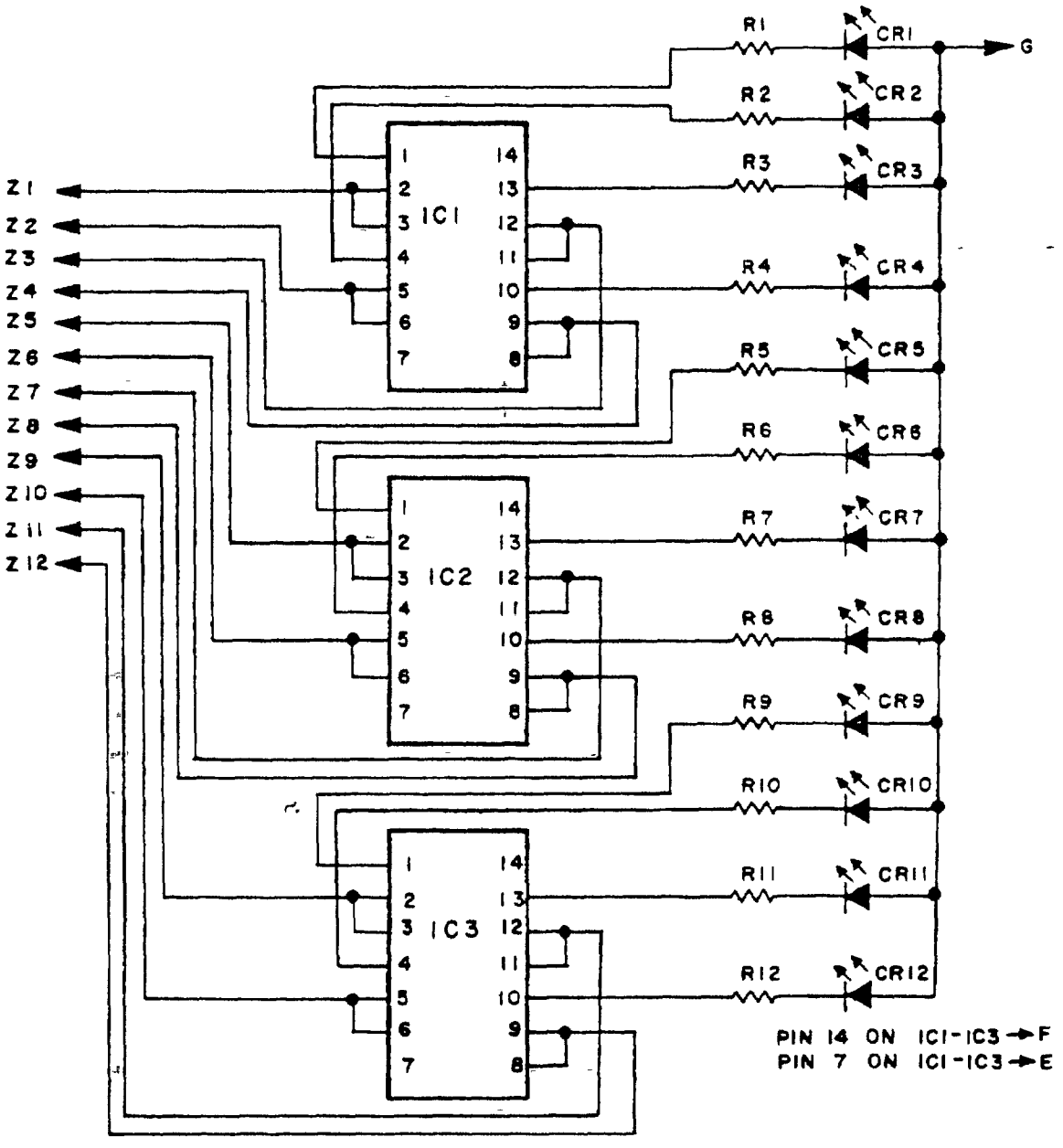



NOTES
1. IC1 = 7404, or equivalent
2. IC2 - IC4 = 7495, or equivalent.

FIGURE 22 Parallel input/parallel output data register



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NOTES:
1. R1 - R12 = 100Ω ±5%, 1/2 W
2. CR1 - CR12 = Light emitting diodes HP 5082-4403, or equivalent.
3. IC1 - IC3 = 7401, or equivalent.
4. Light on = High level
5. Light off = Low level

FIGURE 23 Level detector

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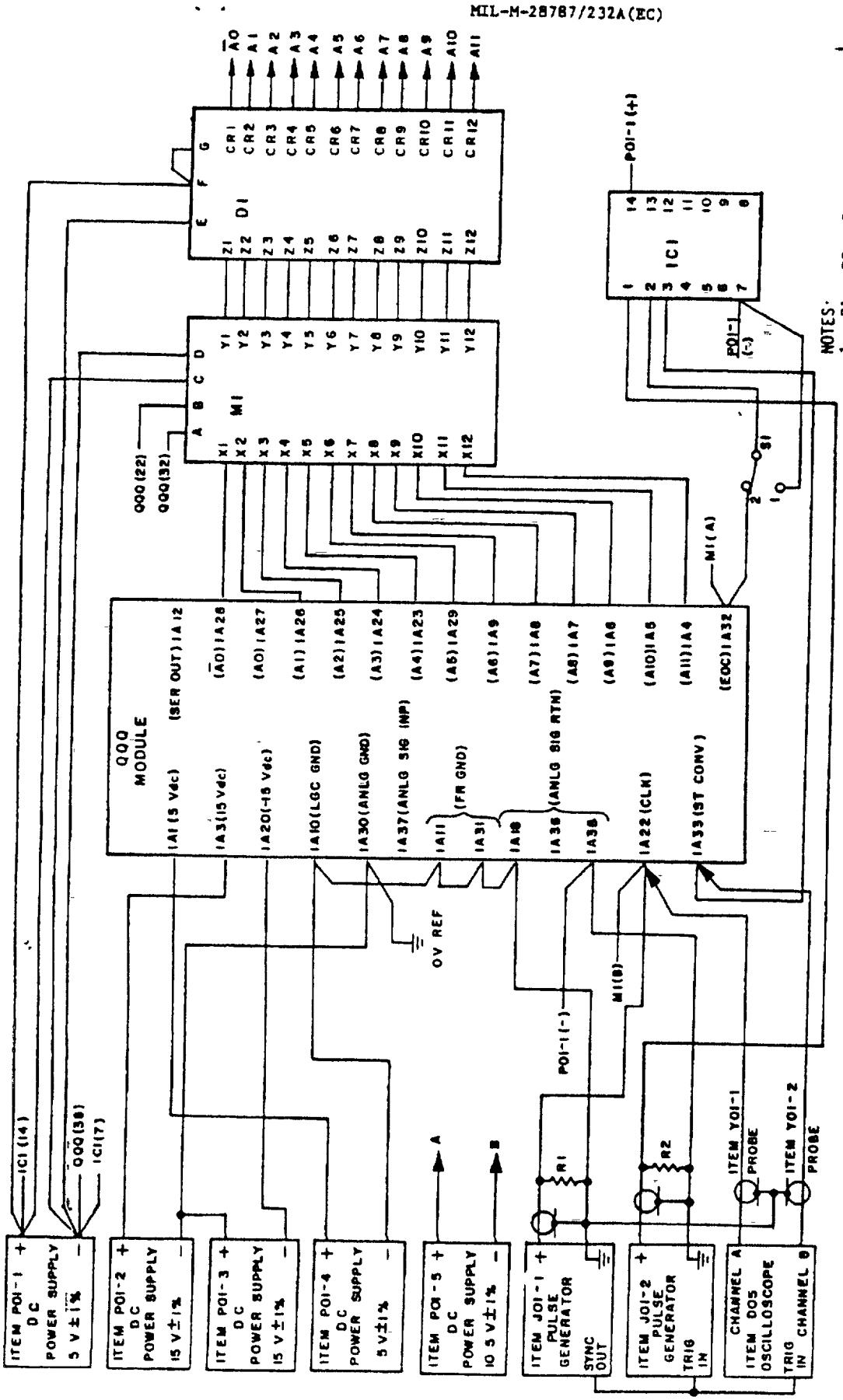
TABLE XVI. Start convert pulse width check test.

| Test condition | Parameter | Connect | | Digital output | | | | | | | | | | | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | $\overline{A0}$ | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 | | |
| P1 | Start convert pulse width check | 37 | 36 | H | L | L | L | L | L | L | L | L | L | L | L | Digital output | 24, 25 |
| P2 | Start convert pulse width check | 36 | 37 | L | H | H | H | H | H | H | H | H | H | H | H | Digital output | 24, 25 |





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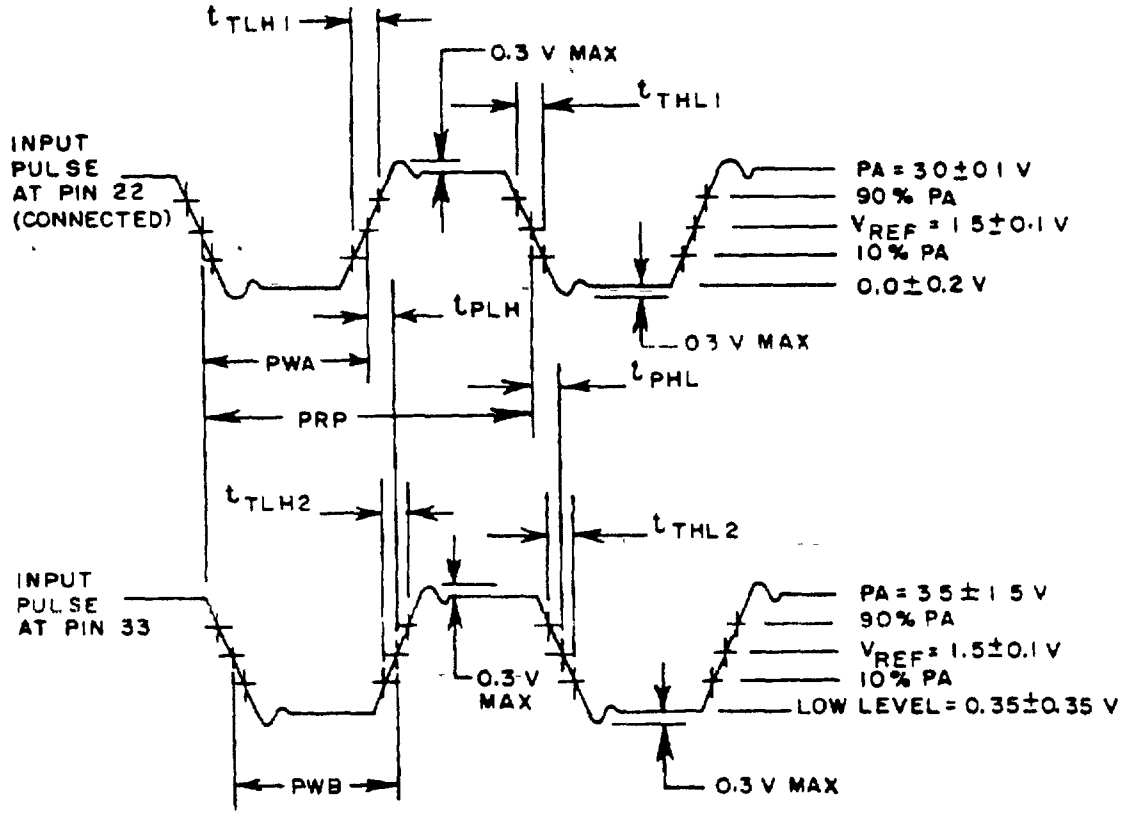



NOTES:
1. R1 = R2 = 51 ±5%, 1/2 W.
2. S1 = SPDT switch.
3. IC1 = 7432, or equivalent.
4. M1 = Data register per figure 22.
5. D1 = Level detector per figure 23.
6. Place switch S1 in position 1 to adjust waveforms per figure 25. Place switch S1 in position 2 to perform test conditions.
7. The digital output displayed by level detectors shall be as specified in table XVI.

FIGURE 24. Start convert pulse width check test circuit for table XVI.



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NOTES
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 200 ±10 ns.
   PRP = 5.0 ±0.5 µs.
   $t_{TLH2} = t_{THL2} = 20 \pm 10$ ns.
   $t_{PLH} = t_{PHL} = 10 \pm 2$ ns.
   PWB = 200 ±10 ns.
2. All voltages are referenced to 0 V unless otherwise specified.

FIGURE 25. Start convert pulse width check test circuit waveforms for table XVI



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TABLE XVII. Power supply noise rejection check test.

| Test condition | Parameter | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| Q1 | P01-2 Power supply noise rejection check | -1.5 | 1.5 | LSB | 26 |
| Q2 | P01-3 Power supply noise rejection check | -1.5 | 1 5 | LSB | 26 |
| Q3 | P01-4 Power supply noise rejection check | -1.5 | 1.5 | LSB | 26 |





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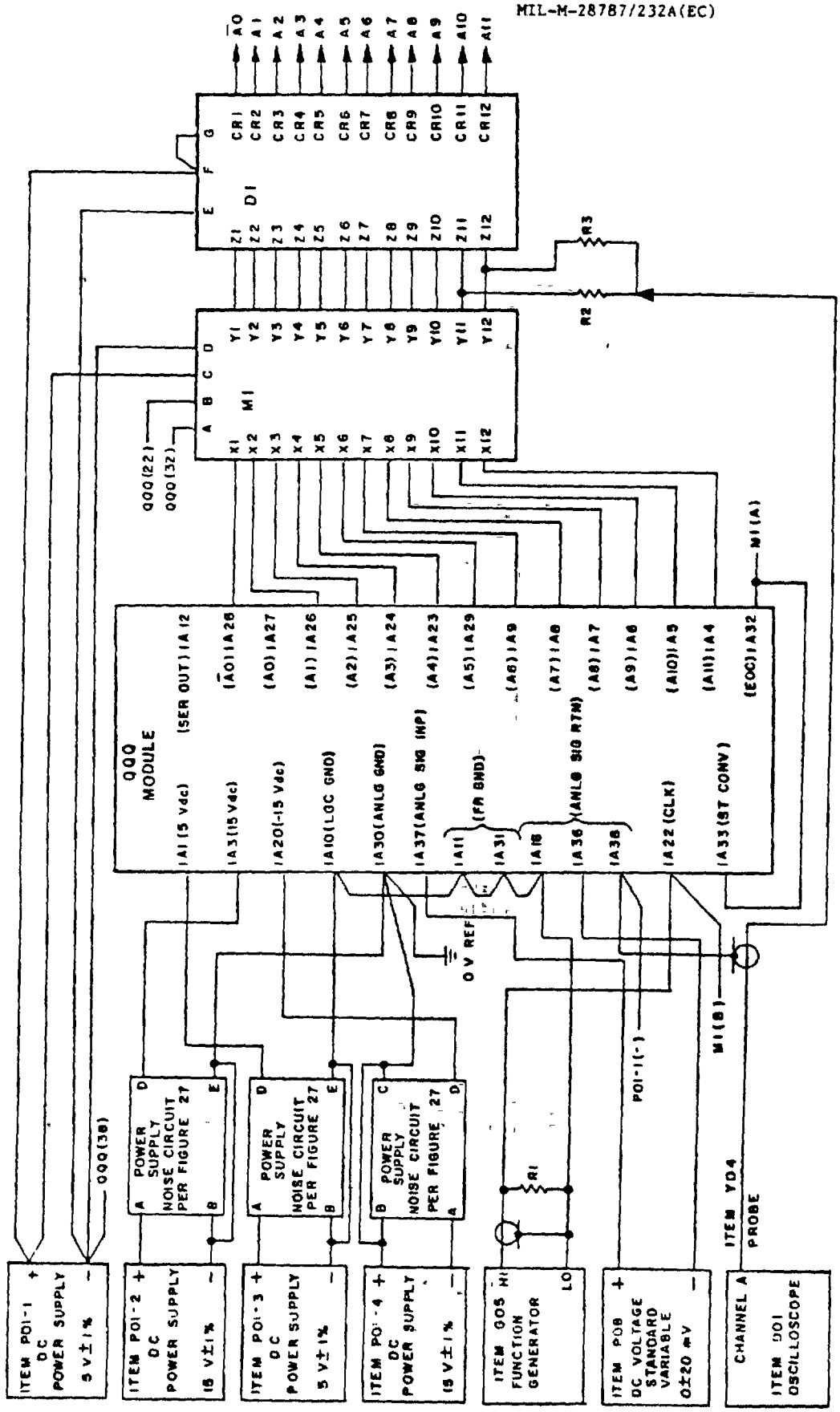



FIGURE 26. Power supply noise rejection check test circuit for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:49Z
Check the source to verify that this is the current version before use.

NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = 4.99 kΩ ±5%, 1/4 W.
3. R3 = 10 kΩ ±5%, 1/4 W.
4. M1 = Data register per figure 22.
5. D1 = Level detector per figure 23.
6. Since the power supply noise circuits are not used simultaneously, only one circuit is required
7. Item G05 setup (set parameters using item D01 referenced to pin 10).
   Square wave output.
   Frequency = 200 kHz ±25%.
   High level = 4.0 ±1.0 V.
   Low level = 0 0 ±0.2 V.
8. Calibration procedure (required only once prior to performing test conditions Q1 - Q3):

   a Connect the equipment as specified on figure 26. Power supply noise circuit shall not be connected at this time.
   b. Set item D01 horizontal display on EXT-X1.
   c. Set item D01 vertical sensitivity to 2V/division.
   d. Adjust item P08 until the digital output (displayed by level detectors) is as indicated below

| $\overline{A0}$ | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| L | L | L | L | L | L | L | L | L | L | L | L |

   e. Adjust the horizontal and vertical controls of item D01 until the display lies on the center vertical graticule and is 2 divisions below the horizontal center point.
   f. Adjust item P08 (0 ±20 mV) until the digital output (displayed by level detectors) just changes to the pattern specified below.

| $\overline{A0}$ | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| H | H | H | H | H | H | H | H | H | H | H | H |

   Record the item P08 setting.
   g. Adjust item P08 (0 ±20 mV) until the digital output (displayed by level detectors) just changes to the pattern specified below:

| $\overline{A0}$ | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| L | L | L | L | L | L | L | L | L | L | L | H |

   Record the item P08 setting.
   h Set item P08 halfway between the readings recorded for steps 8f and 8g. The digital output shall be as specified in 8d.

9 Connect the power supply noise circuit as specified for the appropriate test condition.
10. The item D01 display shall have a 1 to 3 dot display to satisfy the limits of table XVII. If 4 dots are displayed, the module does not satisfy the requirements of table XVII.

FIGURE 26. Power supply noise rejection check test circuit for table XVII - Continued

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Check the source to verify that this is the current version before use.

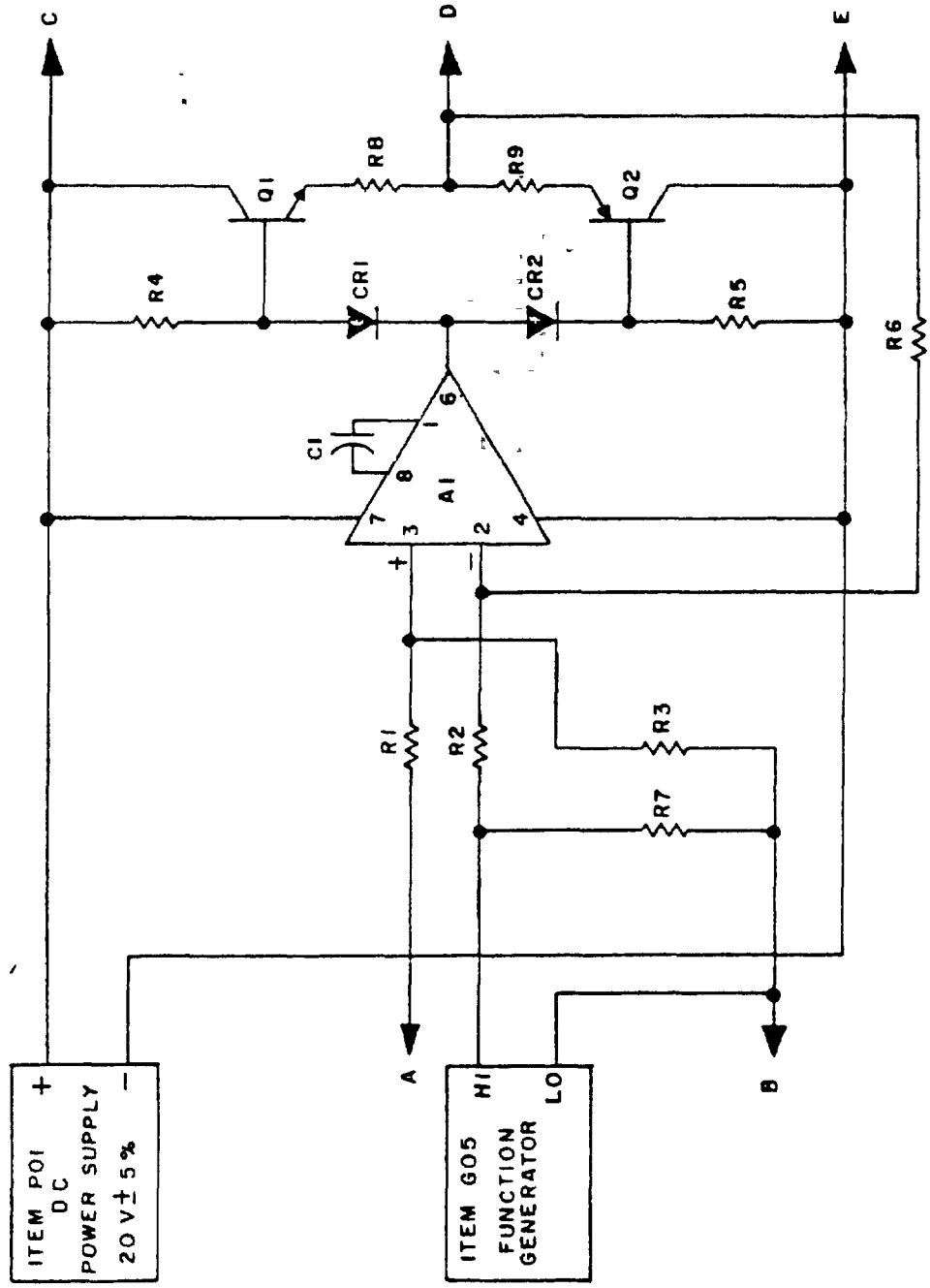



FIGURE 27. Power supply noise circuit.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:49Z
Check the source to verify that this is the current version before use.

NOTES:
1. R1 = R2 = R3 = R6 = 51 kΩ ±1%, 1/8 W.
2. R4 = R5 = 10 kΩ ±5%, 1/4 W.
3. R7 = 51Ω ±5%, 1/2 W.
4. R8 = R9 = 10Ω ±5%, 1/2 W.
5. C1 = 30 pF ±10%, 50 V.
6. CR1 = CR2 = 1N4148, or equivalent
7. Q1 = 2N5150, or equivalent
8. Q2 = 2N5149, or equivalent
9. A1 = 208A, or equivalent
10. Item G05 setup:
    Sine wave output.
    Frequency = 1 kHz ±10%.
    Amplitude = 200 mVrms ±10%.

FIGURE 27. Power supply noise circuit - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:49Z
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TABLE XVIII. Serial output check test

| Test condition | Parameter | Connect | | Serial output | | | | | | | | | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | $\bar{A}0$ | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 | Min | Max | | |
| R1 | Serial output check | 36 | 37 | H | L | H | L | H | L | H | L | H | L | H | L | -3.3398 | -3.3203 | V | 28 |
| R2 | Serial output check | 37 | 36 | L | H | L | H | L | H | L | H | L | H | L | H | 3.3252 | 3.3447 | V | 28 |





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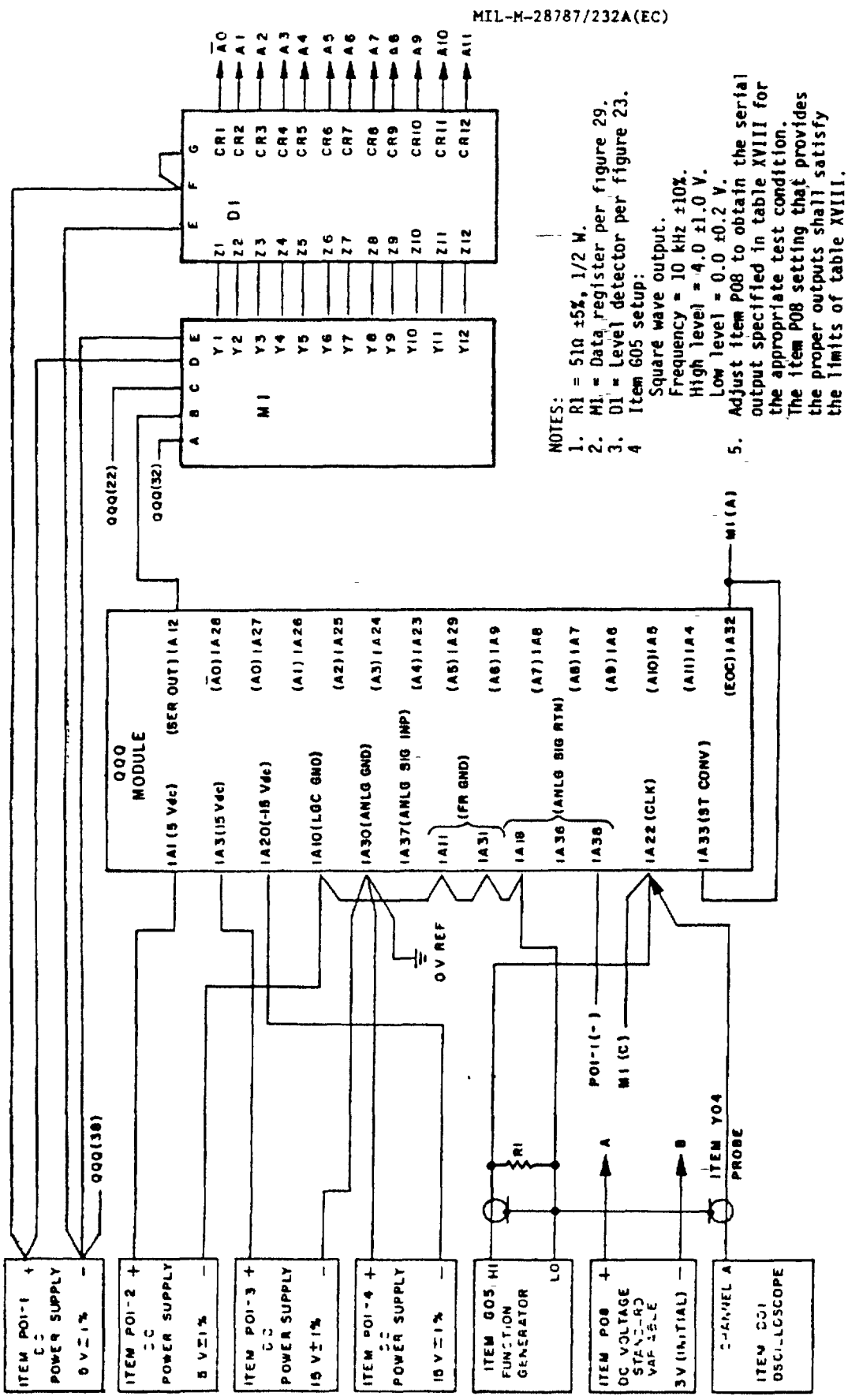
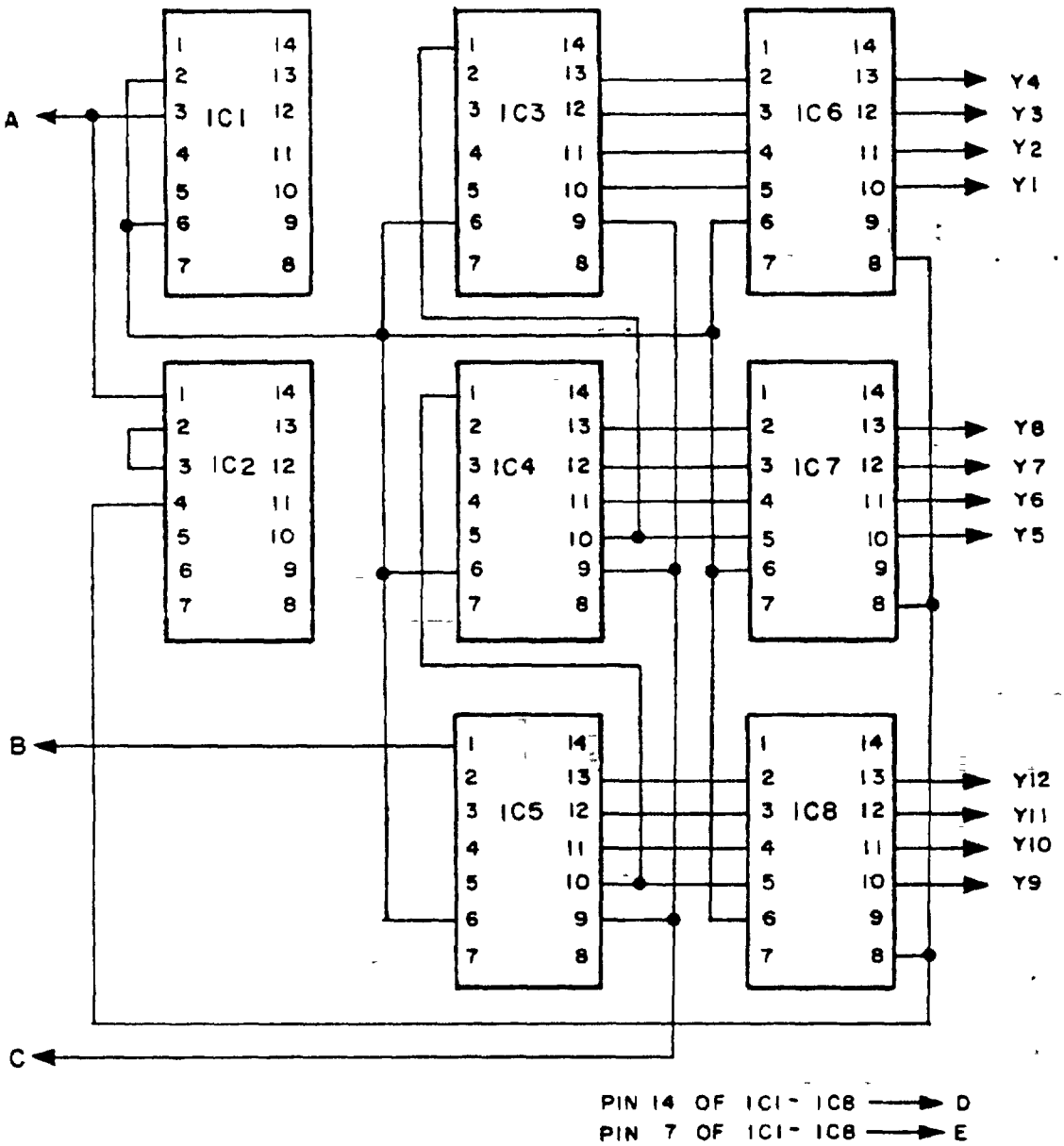



FIGURE 28. Serial output check test circuit for table XVIII.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:49Z
Check the source to verify that this is the current version before use.




NOTES:
1. IC1 = 7474, or equivalent.
2. IC2 = 7404, or equivalent.
3. IC3 - IC8 = 7495, or equivalent

FIGURE 29 Serial input/parallel output to parallel input/parallel output data register.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:49Z
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TABLE XIX. Multiplexed input delay check test.

| Test condition | Parameter | Connect | | Digital output | | | | | | | | | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to point | B to point | $\overline{A0}$ | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 | Min | Max | | |
| S1 | Multiplexed input delay check | D | C | L | L | L | L | L | L | L | L | L | L | L | L | -1 | 1 | LSB | 30 |
| | | | | L | L | L | L | L | L | L | L | L | L | L | H | | | | |
| | | | | L | L | L | L | L | L | L | L | L | L | H | L | | | | |
| S2 | | C | D | L | L | L | L | L | L | L | L | L | L | L | L | | | | |
| | | | | L | L | L | L | L | L | L | L | L | L | L | H | | | | |
| | | | | L | L | L | L | L | L | L | L | L | L | H | L | | | | |
| S3 | | C | D | H | H | H | H | H | H | H | H | H | H | H | L | | | | |
| | | | | H | H | H | H | H | H | H | H | H | H | H | H | | | | |
| | | | | L | L | L | L | L | L | L | L | L | L | L | L | | | | |
| S4 | | D | C | H | H | H | H | H | H | H | H | H | H | H | L | | | | |
| | | | | H | H | H | H | H | H | H | H | H | H | H | H | | | | |
| | | | | L | L | L | L | L | L | L | L | L | L | L | L | | | | |





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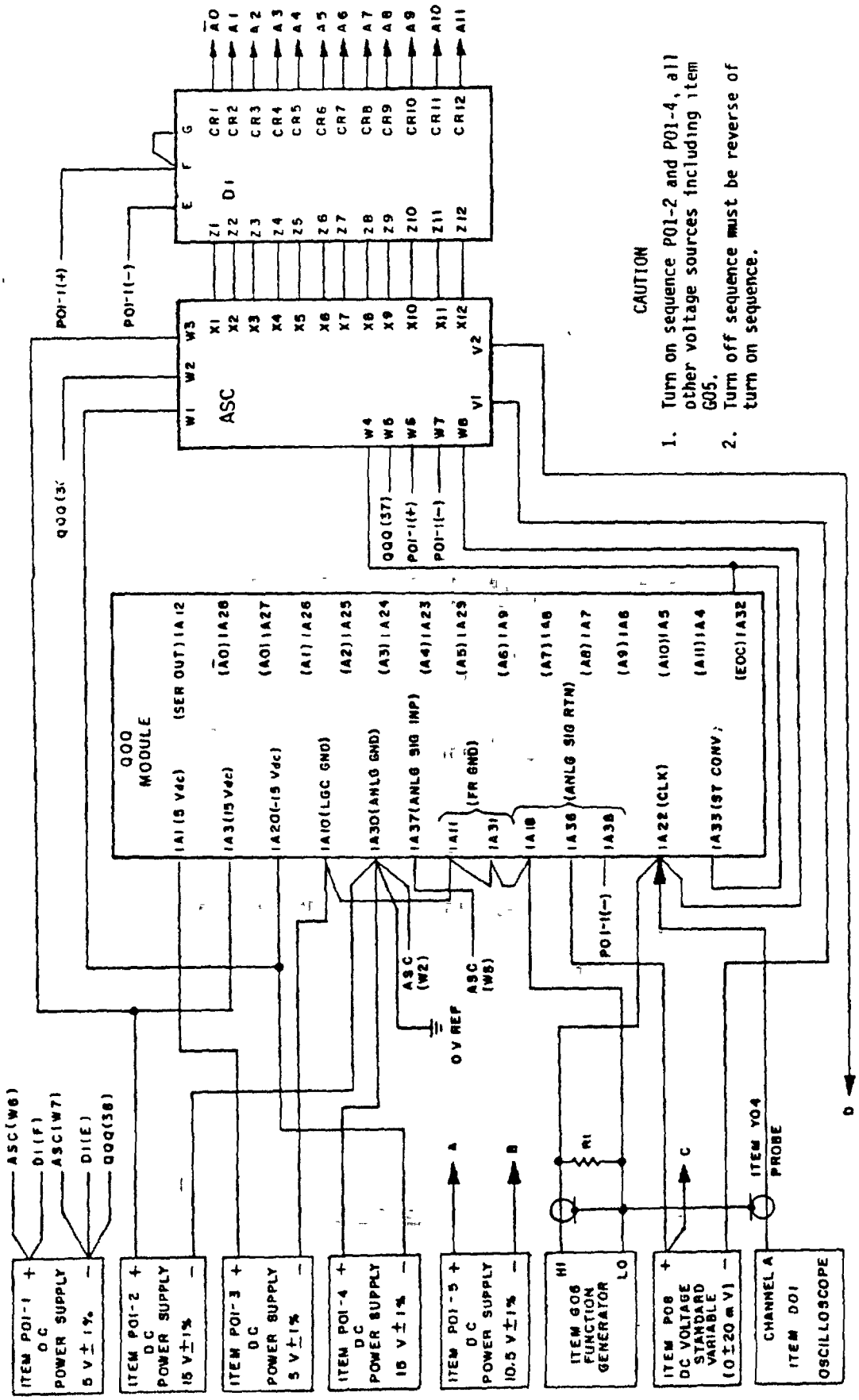



FIGURE 30. Multiplexed input delay check test circuit for table XIX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:49Z
Check the source to verify that this is the current version before use.

NOTES:
1. ASC - Analog input step change circuit per figure 31.
2. D1 - Level detector per figure 23.
3 R1 = 51Ω ±5%, 1/2 W.
4. Item G05 setup:
   Square wave utput
   Frequency = 1 kHz ±10%.
   High level = 4.0 ±1.0 V.
   Low level = 0 ±0.2 V.
5. Test conditions S1 - S2.

   a. Connect the equipment as specified for the appropriate test condition.
   b Adjust item P08 until the digital output code (displayed by level detectors) is as specified below

| $\overline{A0}$ | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| L | L | L | L | L | L | L | L | L | L | L | H |

   c. Increase item G05 frequency to 330 kHz ±5%.
   d. The digital output code displayed by the level detectors shall be one of the three codes as specified in table XIX.

6. Test conditions S3 - S4:

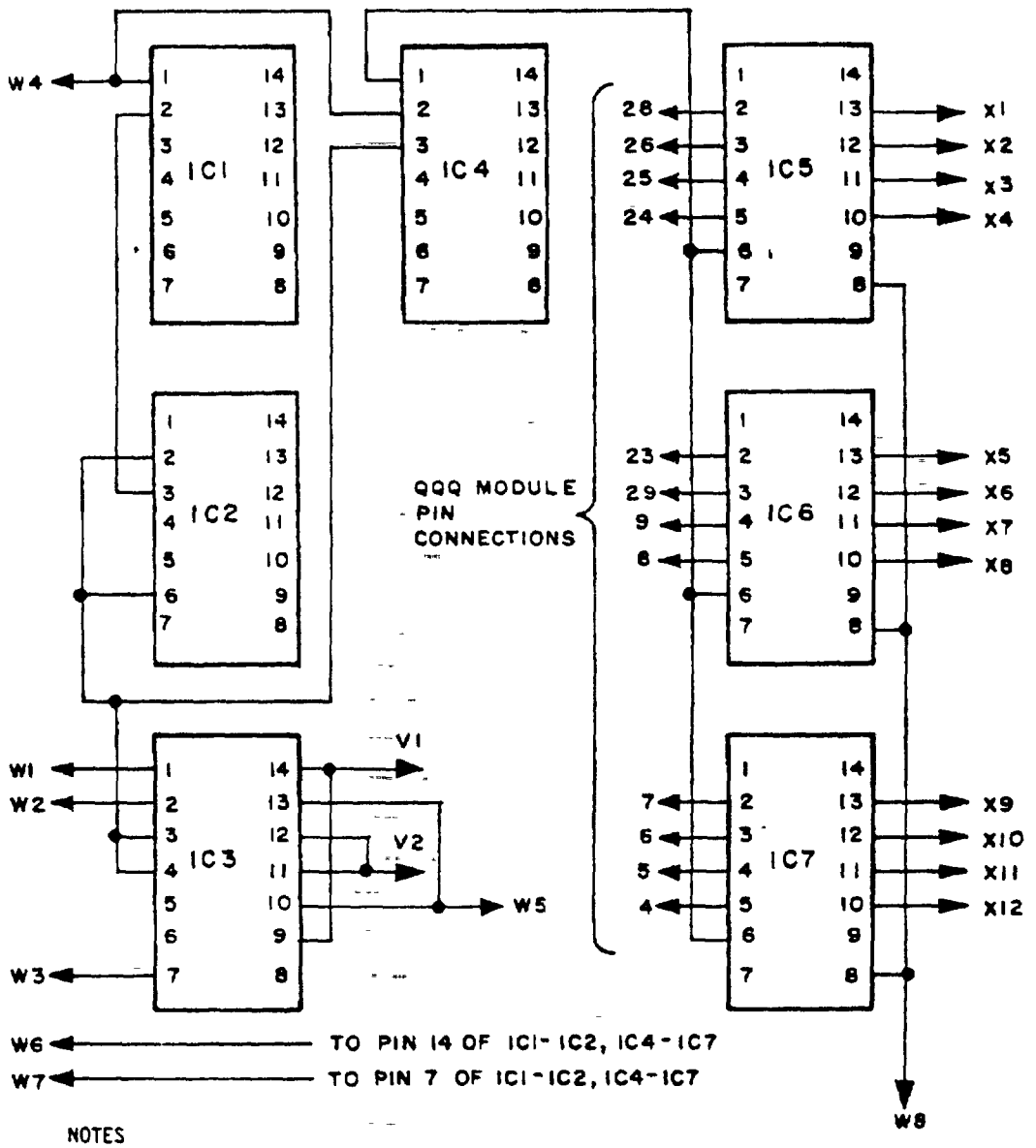
   a Repeat steps 4 and 5a.
   b Adjust item P08 until the digital output code (displayed by level detectors) is as specified below

| $\overline{A0}$ | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| H | H | H | H | H | H | H | H | H | H | H | H |

   c. Repeat steps 5c and 5d.

FIGURE 30. Multiplexed input delay check test circuit for table XIX - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:49Z
Check the source to verify that this is the current version before use.




NOTES
1. IC1 = 7404, or equivalent
2. IC2 = 7474, or equivalent.
3 IC3 = AD7512 KXN, or equivalent.
4 IC4 = 7402, or equivalent.
5. IC5 - IC7 = 7495, or equivalent.

FIGURE 31 Analog input step change circuit.



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*TABLE XX. Serial/parallel output check test

| Test condition | Parameter | Connect | | Digital output | | | | | | | | | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | A0 | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 | Min | Max | | |
| T1 | Serial/parallel output check | 36 | 37 | H | L | H | L | H | L | H | L | H | L | H | L | -3.3398 | -3.3203 | V | 32, 33 |
| T2 | ↓ | 37 | 36 | L | H | L | H | L | H | L | H | L | H | L | H | 3.3252 | 3.3447 | ↓ | ↓ |
| T3 | | 37 | 36 | L | H | L | H | L | H | L | H | L | H | L | H | 3.3252 | 3.3447 | | |
| T4 | | 36 | 37 | H | L | H | L | H | L | H | L | H | L | H | L | -3.3398 | -3.3203 | | |





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Check the source to verify that this is the current version before use.

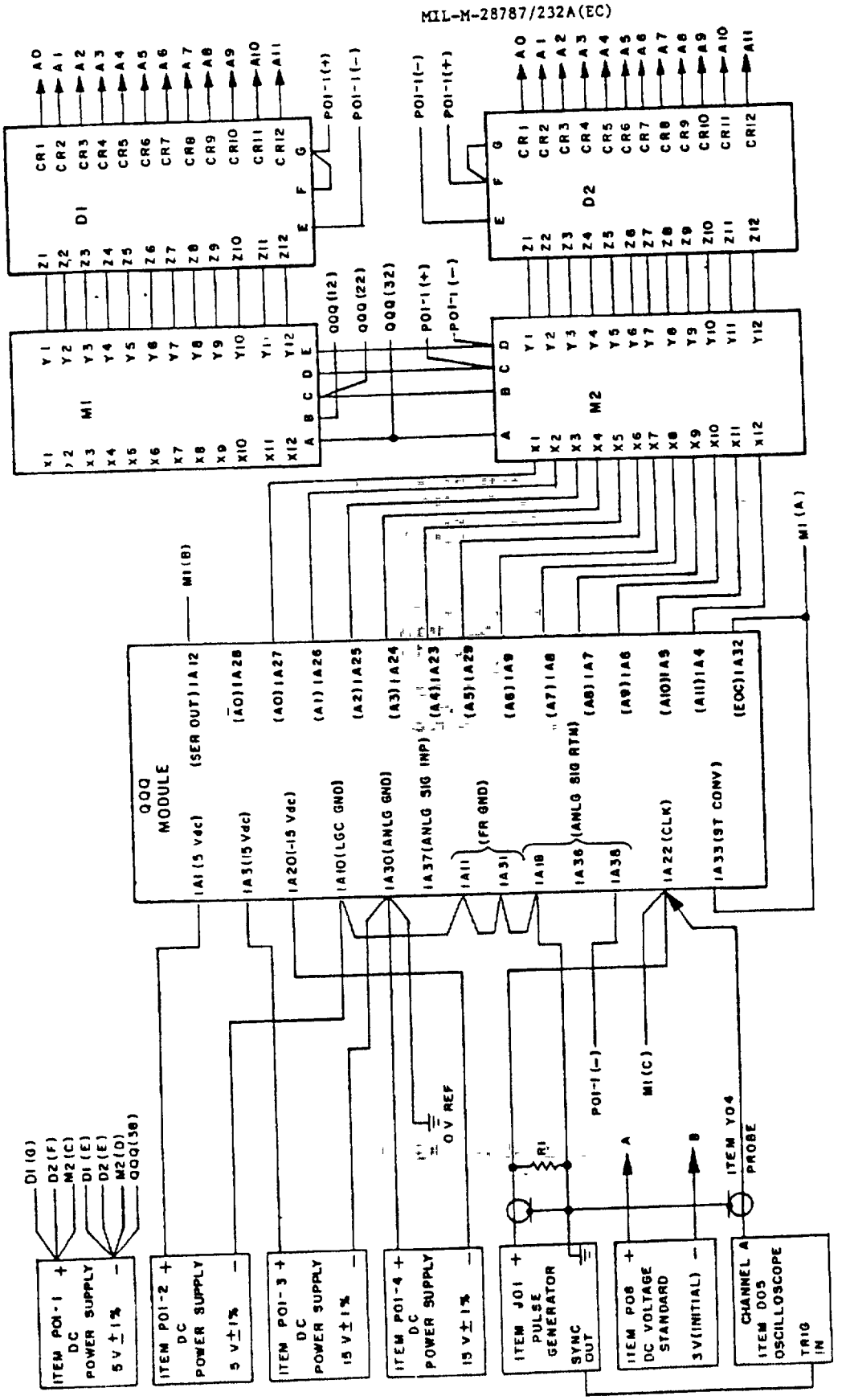



FIGURE 32. Serial/parallel output check test circuit for table XX

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:49Z
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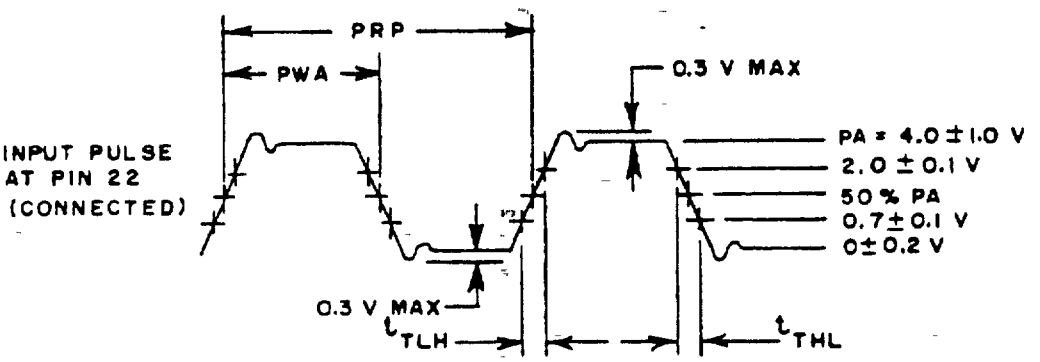
NOTES:
1. M1 - Serial output data register per figure 29.
2. M2 - Parallel output data register per figure 22.
3. D1 - Level detector circuit number 1 per figure 23.
4. D2 - Level detector circuit number 2 per figure 23.
5. R1 = 51Ω ±5%, 1/2 W.
6. Test conditions T1 - T4.

   a. Connect the equipment as specified for the appropriate test conditions.
   b. Adjust item P08 within the limits of table XX to obtain the digital output code (displayed by level detector circuit #1) specified in table XX for the appropriate test condition.
   c. The digital output code displayed by level detector circuit #2 shall be the same as that displayed by level detector circuit #1. If the digital codes are not the same, the module shall be considered failed.

FIGURE 32. Serial/parallel output check test circuit for table XX - Continued



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:49Z
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NOTES:
1. Input pulse parameters for test conditions T1 and T2.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PWA = 2.5 ±0.25 µs.
   PRP = 5 ±0.5 µs.
2. Input pulse parameters for test conditions T3 and T4.
   $t_{TLH} = t_{THL} = 100 \pm 10$ ns.
   PWA = 2.5 ±0.25 µs.
   PRP = 5 ±0 5 µs.
3. All voltages are referenced to 0 V unless otherwise specified.

FIGURE 33 Serial/parallel output waveforms for table XX.



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TABLE XXI. Hysteresis test.

| Test condition | Parameter | DC voltage standard setting ± 0.04 V | Digital output code | | | | | | | | | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | $\bar{A}0$ | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 | Min | Max | | |
| U1 | Hysteresis | 7.50 V | H | L | H | L | L | L | L | L | L | L | L | L | -2 44 | 2.44 | mV | 34 |
| U2 | | 5.00 V | H | H | L | L | L | L | L | L | L | L | L | L | -2.44 | 2,44 | | |
| U3 | | 2.50 V | H | H | H | L | L | L | L | L | L | L | L | L | -2.44 | 2.44 | | |
| U4 | | 0.00 V | L | L | L | L | L | L | L | L | L | L | L | L | -4.88 | 4.88 | | |
| U5 | | -2.50 V | L | L | H | L | L | L | L | L | L | L | L | L | -2.44 | 2.44 | | |
| U6 | | -5.00 V | L | H | L | L | L | L | L | L | L | L | L | L | -2.44 | 2.44 | | |
| U7 | | -7.50 V | L | H | H | L | L | L | L | L | L | L | L | L | -2.44 | 2.44 | | |





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TABLE XXII. Differential nonlinearity test.

| Test Cond | Parameter | Leading edge | Trailing edge | DC voltage standard setting ± 0.04 V | Digital output code | | | | | | | | | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | $\bar{A}0$ | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 | Min | Max | | |
| V1 | Differential nonlinearity | E5 | E8 | 4.88 mV | H | H | H | H | H | H | H | H | H | H | H | H | 0.50 | 9.76 | mV | 34 |
| V2 | | E7 | E2 | 0.0 V | L | L | L | L | L | L | L | L | L | L | L | L | | | | |
| V3 | | E1 | E4 | -4.88 mV | L | L | L | L | L | L | L | L | L | L | L | H | | | | |
| V4 | | E3 | E6 | -9.76 mV | L | L | L | L | L | L | L | L | L | L | H | L | | | | |
| V5 | | E7 | E2 | -19.52 mV | L | L | L | L | L | L | L | L | L | H | L | L | | | | |
| V6 | | | | -39.04 mV | L | L | L | L | L | L | L | L | H | L | L | L | | | | |
| V7 | | | | -78.08 mV | L | L | L | L | L | L | L | H | L | L | L | L | | | | |
| V8 | | | | -156.16 mV | L | L | L | L | L | L | H | L | L | L | L | L | | | | |
| V9 | | | | -312.32 mV | L | L | L | L | L | H | L | L | L | L | L | L | | | | |
| V10 | | | | -625 mV | L | L | L | L | H | L | L | L | L | L | L | L | | | | |
| V11 | | | | -1.25 V | L | L | L | H | L | L | L | L | L | L | L | L | | | | |
| V12 | | | | -2.50 V | L | L | H | L | L | L | L | L | L | L | L | L | | | | |
| V13 | | | | -5.00 V | L | H | L | L | L | L | L | L | L | L | L | L | | | | |
| V14 | | E5 | E8 | 4 88 mV | H | H | H | H | H | H | H | H | H | H | H | H | | | | |
| V15 | | E7 | E2 | -0.0 V | L | L | L | L | L | L | L | L | L | L | L | L | | | | |





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TABLE XXII. Differential nonlinearity test - Continued.

| Test Cond | Parameter | Leading edge | Trailing edge | DC voltage standard setting ±0.04 V | Digital output code | | | | | | | | | | | | Initial limits 25° +5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | $\bar{A}0$ | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 | Min | Max | | |
| V16 | Differential nonlinearity | E1 | E4 | -4.88 mV | L | L | L | L | L | L | L | L | L | L | L | H | 0.50 | 9.76 | mV | 34 |
| V17 | ↓ | E3 | E6 | -9.76 mV | L | L | L | L | L | L | L | L | L | L | H | L | ↓ | ↓ | ↓ | ↓ |
| V18 | | E7 | E2 | -19.52 mV | L | L | L | L | L | L | L | L | L | H | L | L | | | | |
| V19 | | ↓ | ↓ | -39.04 mV | L | L | L | L | L | L | L | L | H | L | L | L | | | | |
| V20 | | | | -78.08 mV | L | L | L | L | L | L | L | H | L | L | L | L | | | | |
| V21 | | | | -156.16 mV | L | L | L | L | L | L | H | L | L | L | L | L | | | | |
| V22 | | | | -312.32 mV | L | L | L | L | L | H | L | L | L | L | L | L | | | | |
| V23 | | | | -625 mV | L | L | L | L | H | L | L | L | L | L | L | L | | | | |
| V24 | | | | -1.25 V | L | L | L | H | L | L | L | L | L | L | L | L | | | | |
| V25 | | | | -2.50 V | L | L | H | L | L | L | L | L | L | L | L | L | | | | |
| V26 | ↓ | ↓ | ↓ | -5.00 V | L | H | L | L | L | L | L | L | L | L | L | L | ↓ | ↓ | ↓ | ↓ |





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* TABLE XXIII. Linearity error test.

| Test cond | Parameter | Step trailing edge | DC voltage standard setting ± 0.02 V | Digital output code | | | | | | | | | | | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | $\bar{A}0$ | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 | Min | Max | | |
| W1 | Linearity error | E1 | 7.50 V | H | L | H | L | L | L | L | L | L | L | L | L | 7.4976 | 7.5024 | V | 34 |
| W2 | | | 5.00 V | H | H | L | L | L | L | L | L | L | L | L | L | 4.9976 | 5.0024 | | |
| W3 | | | 2.50 V | H | H | H | L | L | L | L | L | L | L | L | L | 2.4976 | 2.5024 | | |
| W4 | | | 0.00 V | L | L | L | L | L | L | L | L | L | L | L | L | -0.0024 | 0.0024 | | |
| W5 | | | -2.50 V | L | L | H | L | L | L | L | L | L | L | L | L | -2.4976 | -2.5024 | | |
| W6 | | | -5.00 V | L | H | L | L | L | L | L | L | L | L | L | L | -4.9976 | -5.0024 | | |
| W7 | | | -7.50 V | L | H | H | L | L | L | L | L | L | L | L | L | -7.4976 | -7.5024 | | |





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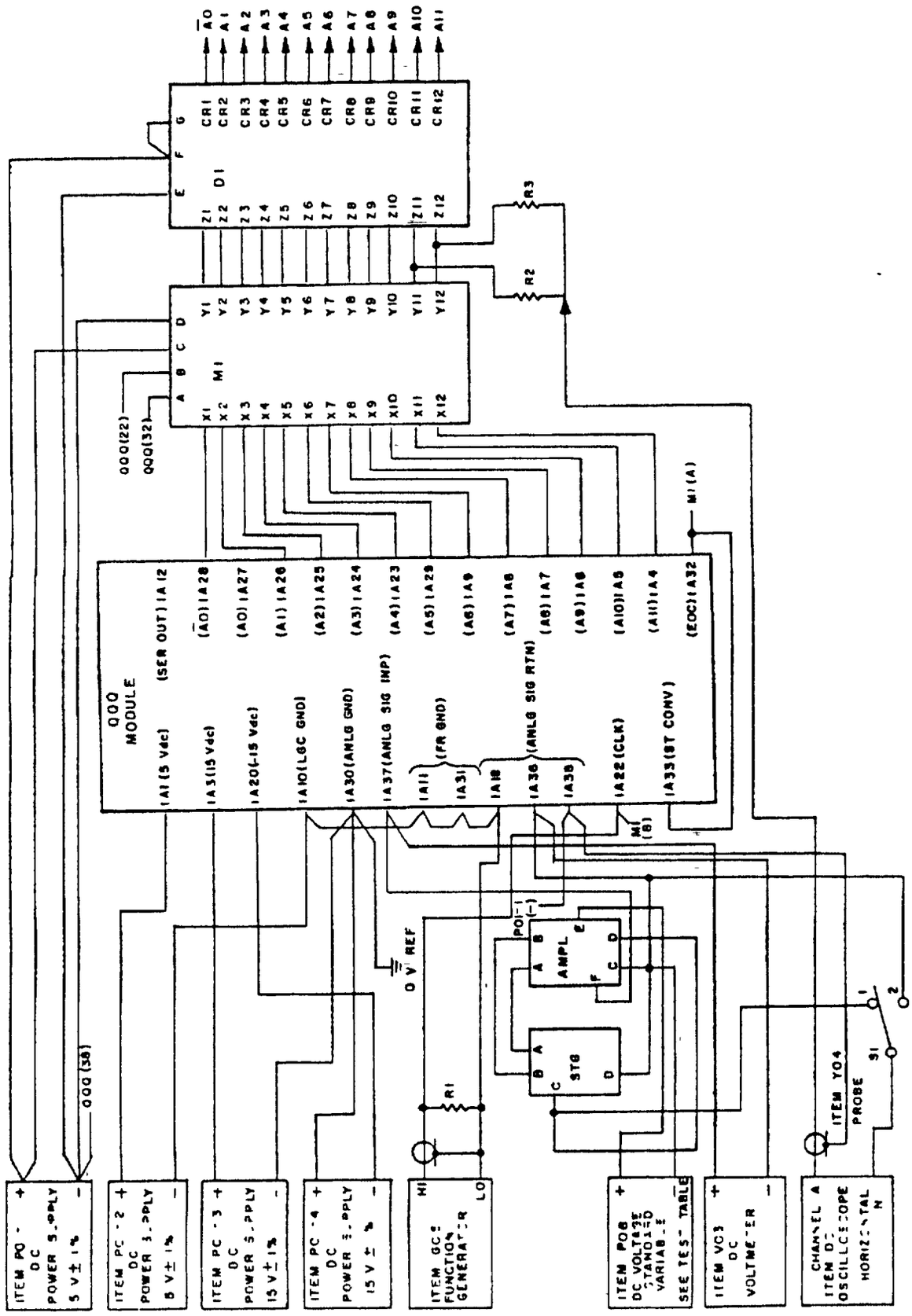



FIGURE 34. Hysteresis, differential nonlinearity, and linearity test circuit for tables XXI, XXII, and XXIII - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:49Z
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NOTES
1. M1 - Data register per figure 22.
2. D1 - Level detector per figure 23.
3. AMPL - Summing amplifier per figure 36.
4. STG - Sawtooth dither generator per figure 35
5. R1 - 51Ω ±5%, 1/2 W
6. R2 - 4.99 kΩ ±5%, 1/4 W.
7. R3 - 10 kΩ ±5%, 1/4 W.
8. S1 - SPDT.
9. Item G05 setup:
   Square wave output.
   Frequency = 330 kHz ±5%.
   High level = 4.0 ±1.0 V.
   Low level = 0 ±0.2 V.
   Set parameters using item D01, referenced to pin 10.
10. Calibration procedure for test conditions U1 - U7.

    a. Set horizontal display of item D01 to EXT-X1 and the vertical sensitivity to 2 V/division.
    b Set item P08 to zero volts and switches S1, S2, S3, and S4 to position 1. Adjust vertical position control until display is centered.
    c Adjust horizontal variable control of item D01 until at least 50% of the steps displayed are 1 ±0.5 divisions wide
    d. Set switch S3 to position 2. The voltage reading on item V03 shall be 0 0 ±1.0 mV. If this criteria is not met, replace A1 of figure 36 until criteria is satisfied
    e Set switch S3 to position 1 and switch S1 to position 2
    f Adjust the horizontal position control of item D01 until display is alined (±0.1 division) with the center vertical graticule.
    g. Set switch S1 to position 1. The display shall be as shown on figure 37.

11 Test conditions U1 - U7.

    a. Set switches S1, S2, and S4 to position 1 and switch S3 to position 2.
    b Set item P08 per table XXI to obtain the digital output code, displayed by level detectors, specified for the appropriate test condition.
    c Set switch S3 to position 1 and adjust item P08 until E7, defined in figure 37, closest to the center vertical graticule is alined (±0.1 divisions) with the graticule. Record the setting of item P08
    d. Set switches S2 and S3 to position 2 and repeat steps 11b thru 11c
    e. The difference in the recorded values of steps 11c and 11d shall satisfy the requirements of table XXI.
    f. Repeat steps 11a and 11b.
    g. Set switch S3 to position 1 and adjust item P08 until E2, defined in figure 37, closest to the center vertical graticule is alined (±0 1 divisions) with the graticule. Record the setting of item P08.
    h. Set switches S2 and S3 to position 2 and repeat steps 11b and 11g.
    i. The difference in the recorded values of steps 11g and 11h shall satisfy the limits of table XXI.

FIGURE 34 Hysteresis, differential nonlinearity, and linearity test circuit for tables XXI, XXII, and XXIII - Continued.

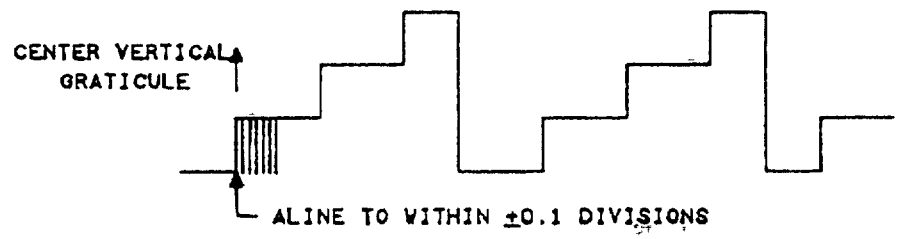


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*12. Calibration procedure for test conditions V1-V26 and W1-W7, must be performed each time a new module is tested.

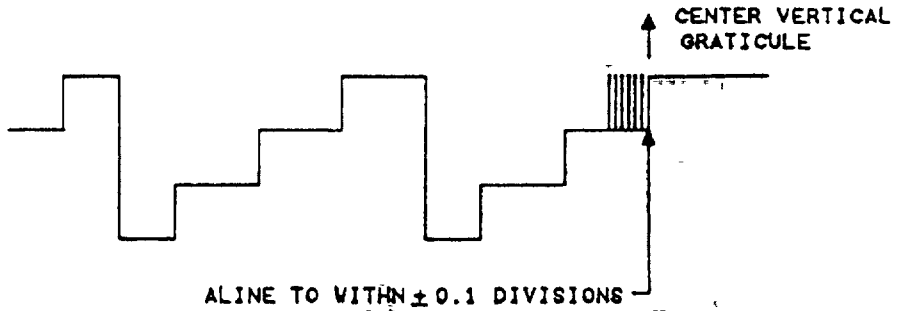
a Set horizontal display of item D01 to EXT-X1 and the vertical sensitivity to 2 V/division.
b Set item P08 to zero volts and switches S1, S2, S3, and S4 to position 1. Adjust vertical position control until display is centered.
c. Adjust horizontal variable control of item D01 until at least 50% of the steps displayed are 1 ±0.5 divisions wide.
d Set switch S3 to position 2. The voltage reading on item V03 shall be 0.0 ±1.0 mV. If this criteria is not met, replace A1 of figure 36 until this criteria is satisfied.

e. Set switch S3 to position 1 and switch S4 to position 2.
f. Aline E1 of the display, defined in figure 37, closest to the center vertical graticule with the graticule (±0.1 divisions) by adjusting the horizontal position control.
g Set item P08 to +10 V and adjust R8 of figure 36 until the positive saturation display as defined below is obtained.




Positive saturation oscilloscope display

h Set item P08 to -9 9902 Vdc. Adjust the horizontal position control until the negative saturation display is alined as shown below




i Repeat steps g and h at least once or as many times as required to obtain both alinements within ±0 1 divisions

FIGURE 34 Hysteresis, differential nonlinearity, and linearity test circuit for tables XXI, XXII, and XXIII



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13. Test conditions V1 - V13.

a Set switches S2, S3, and S4 to position 2 and switch S1 to position 1.
b Set item P08 per table XXII to obtain the digital output code (displayed by level detectors) specified for the appropriate test condition
c. Set switch S3 to position 1. Vary item P08 until the step leading edge as defined in table XXII and in figure 37, closest to the center vertical graticule is alined (±0.1 divisions) with the graticule. Record setting of item P08.
d Adjust item P08 until the step trailing edge as defined in table XXII and in figure 37, closest to the center vertical graticule is alined (±0.1 divisions) with the graticule. Record the setting of item P08.
e. The difference in the recorded values of steps 13c and 13d shall satisfy the limits of table XXII.

14 Test conditions V14 - V26.

a Set switches S1 and S2 to position 1 and switches S3 and S4 to position 2.
b. Repeat steps 13b thru 13e.

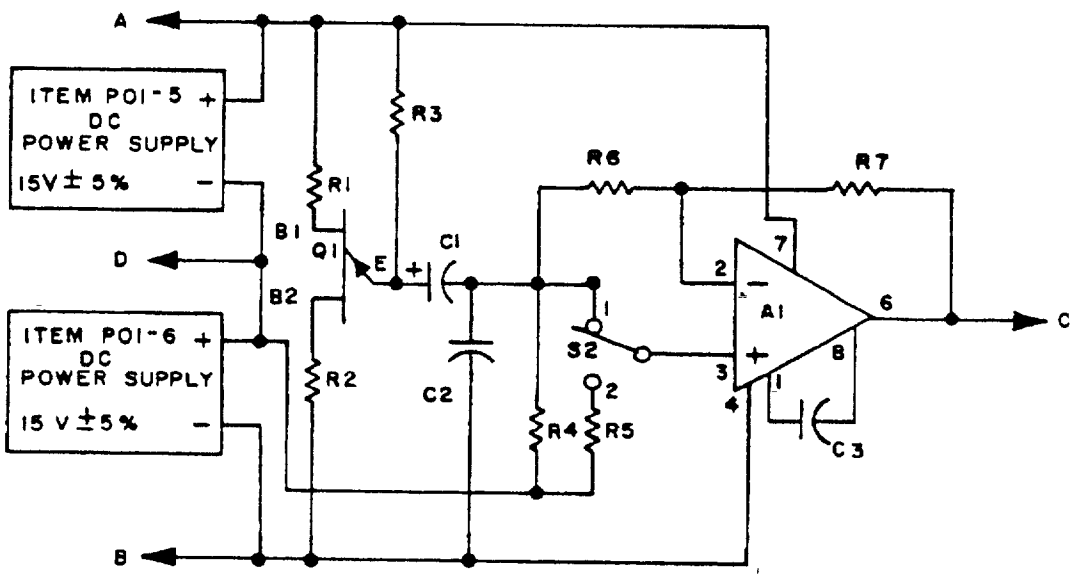
* 15. Test conditions W1 - W7.

a Recalibrate test setup per steps 12a - 12i
b. Set switches S2, S3, and S4 to position 2 and switch S1 to position 1.
c Set item P08 per table XXIII to obtain the digital output code (displayed by level detectors) specified for the appropriate test condition.
d. Set switch S3 to position 1. Vary item P08 until the trailing edge of the step to be measured, defined in figure 37, closest to the center vertical graticule is alined (±0.1 divisions) with the graticule.
e. The settling of item P08 shall satisfy the limits of table XXIII

FIGURE 34 Hysteresis, differential nonlinearity, and linearity test circuit for tables XXI, XXII, and XXIII - Continued



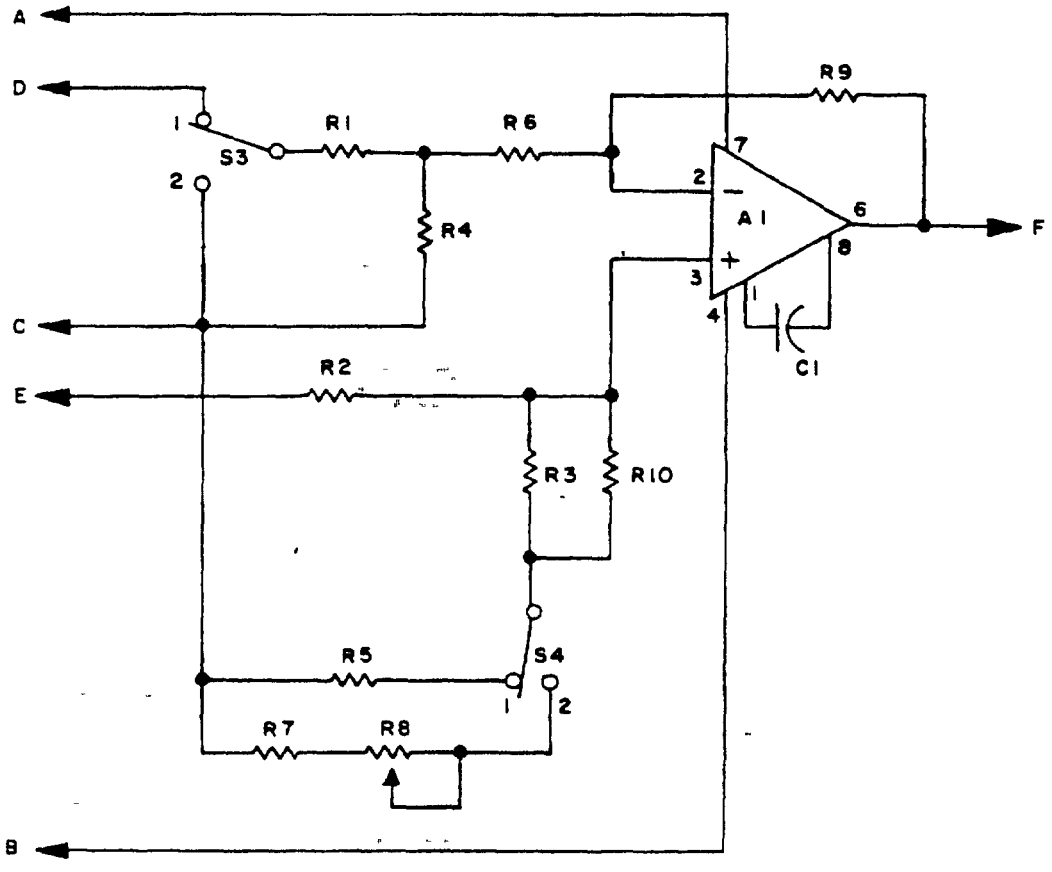
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NOTES:
1. R1 = 100Ω ±5%, 1/4 W.
2. R2 = 10Ω ±5%, 1/4 W.
3. R3 = Select for an oscillation period of 50 ms ±10% from point D to point E.
4. R4 = R6 = R7 = 200 kΩ ±1%, 1/10 W.
5. R5 = 133 kΩ ±1%, 1/10 W.
6. C1 = 1 µF ±10%, 50 Vdc.
7. C2 = 0.47 µF ±10%, 50 Vdc.
8. C3 = 33 pF ±10%.
9. Q1 = 2N491B. or equivalent.
* 10. A1 = LF256, OP-16F, or equivalent
11. S2 = SPDT switch.

FIGURE 35 Dither (sawtooth) generator

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NOTES:
1. R1, R2, R5, R6, R9, R10 = 10 kΩ ±0.01%, 1/10 W.
2. R3, R4 = 75Ω ±0.01%, 1/10 W.
3. R7 = 9.76 kΩ ±1%, 1/10 W.
4. R8 = 2 kΩ, variable, 1/10 W.
5. C1 = 33 pF ±10%.
* 6. A1 = OP-16F, or equivalent.
7. S3, S4 = SPDT switch.

FIGURE 36. Summing amplifier.



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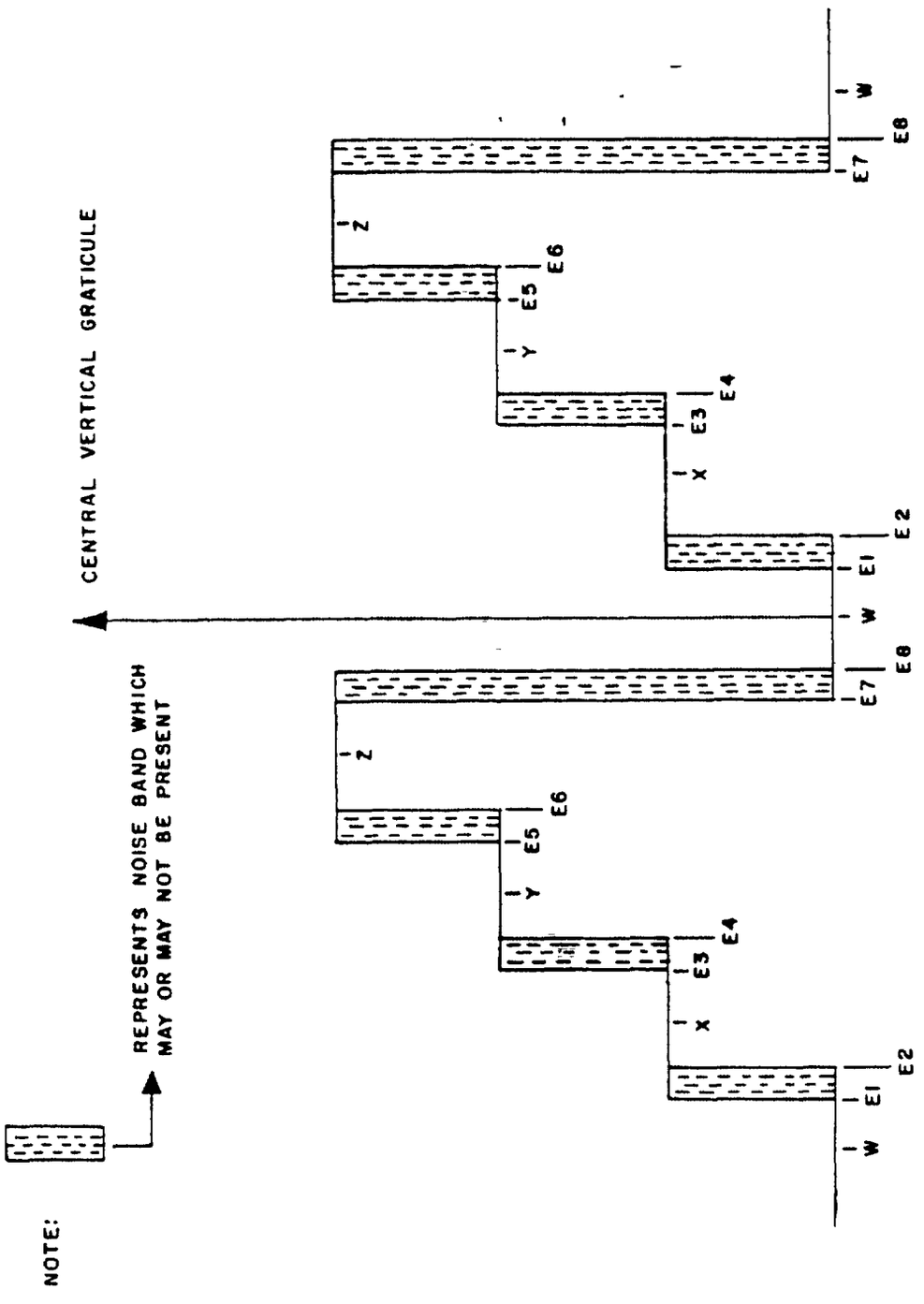



FIGURE 37. Oscilloscope display



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TABLE XXIV. Zero error test.

| Test condition | Parameter | DC Power supply settings ± 1% | | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | PQ1-3 | PQ1-4 | Min | Max | | |
| X1 | Zero error | 15.75 V | 15.75 V | -4.88 | 4.88 | mV | 38 |
| X2 | ↓ | 14.25 V | 15.75 V | ↓ | ↓ | ↓ | ↓ |
| X3 | ↓ | 14.25 V | 14.25 V | ↓ | ↓ | ↓ | ↓ |
| X4 | ↓ | 15.75 V | 14.25 V | ↓ | ↓ | ↓ | ↓ |





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TABLE XXV. Absolute error test.

| Test condition | Parameter | DC Power supply settings ± 1% | | Initial limits 25° +5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | PO1-3 | PO1-4 | Min | Max | | |
| Y1 | Absolute error | 15.75 V | 15 75 V | 9.9776 | 10.0176 | V | 38 |
| Y2 | ↓ | 14.25 V | 15.75 V | ↓ | ↓ | ↓ | ↓ |
| Y3 | | 14.25 V | 14.25 V | | | | |
| Y4 | | 15.75 V | 14 25 V | | | | |
| Y5 | | 15.75 V | 15 75 V | -9.9727 | -10.0127 | | |
| Y6 | | 14.25 V | 15.75 V | | | | |
| Y7 | | 14.25 V | 14.25 V | | | | |
| Y8 | | 15.75 V | 14.25 V | | | | |





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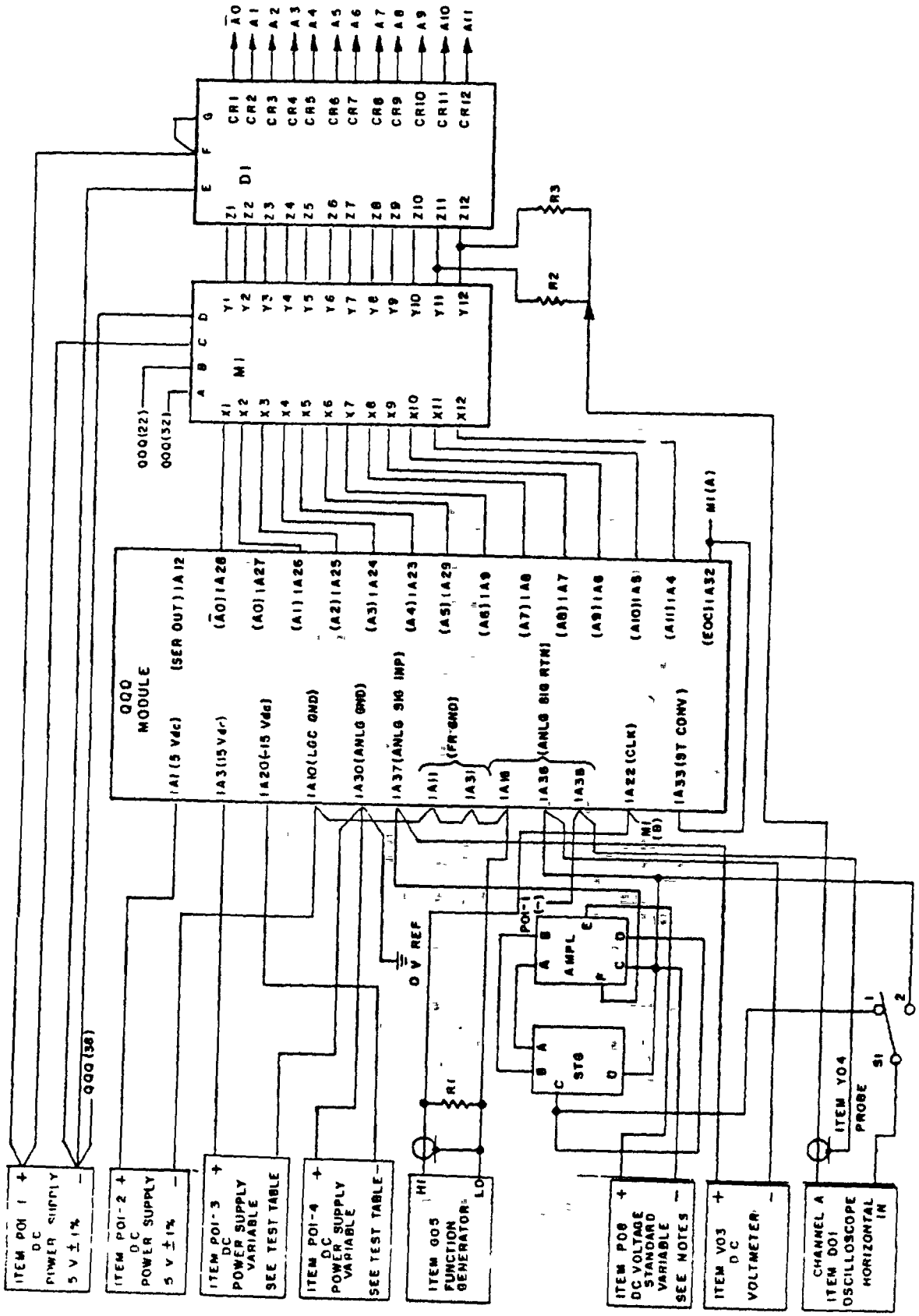



FIGURE 38. Zero error and absolute error test circuit for tables XXIV and XXV.



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NOTES:
1. M1 - Data register per figure 22.
2. D1 - Level detector per figure 23.
3. AMPL - Summing amplifier per figure 36.
4. STG - Sawtooth dither generator per figure 35.
5. R1 - 51Ω ±5%, 1/2 W.
6. R2 - 4.99 kΩ ±5%, 1/4 W.
7. R3 - 10 kΩ ±5%, 1/4 W
8. S1 - SPDT switch.
9. Item G05 setup:
   Square wave output
   Frequency = 200 kHz ±10%.
   High level = 4.0 ±1.0 V.
   Low level = 0.0 ±0.2 V.
   Set parameters using item D01, referenced to pin 10.
10. Calibration procedure for test conditions X1 - X4 and Y1 - Y8.

    a. Set horizontal display of item D01 to EXT-X1 and the vertical sensitivity to 2 V/division
    b. Set item P08 to zero volts and switches S1, S2, S3, and S4 to position 1. Adjust vertical position control to center the display.
    c. Adjust horizontal variable control of item D01 until at least 50% of the steps displayed are 1 ±0 5 divisions wide.
    d. Set switch S3 to position 2. The voltage reading on item V03 shall be 0.0 ±1.0 mV. If this criteria is not met, replace A1 of figure 36 until criteria is satisfied.
    e. Set switch S3 to position 1 and switch S1 to position 2.
    f. Adjust the horizontal position control of item D01 until the dot display is alined (±0.1 divisions) with the center vertical graticule.
    g. Set switch S1 to position 1. The display shall be as shown on figure 37

11. Test conditions X1 - X4.

    a Set switches S2 and S3 to position 2 and switches S1 and S4 to position 1
    b Adjust item P08 until the voltage reading on item V03 is 0.0 ±0 1 mV Record the setting of item P08.
    c. Adjust item P08, 0.0 ±20 mV, to obtain the digital output code, displayed by level detectors, specified below:

| $\overline{A0}$ | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| L | L | L | L | L | L | L | L | L | L | L | L |

    d Set switch S3 to position 1 and adjust item P08 until "W", as defined on figure 37, closest to the center vertical graticule is alined (±0.1 divisions) with the graticule. Record the setting of item P08.
    e. The difference in the recorded values of steps 11b and 11d shall satisfy the limits of table XXIV

FIGURE 38 Zero error and absolute error test circuit for tables XXIV and XXV - Continued.



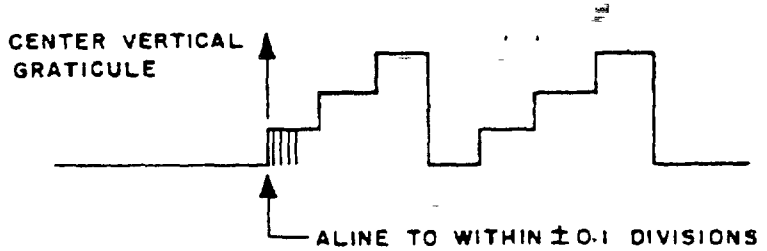
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12. Test conditions Y1 - Y4.

a. Set switches S2 and S3 to position 2 and switches S1 and S4 to position 1
b Adjust item P08 to 9.995 ±0.04 V to obtain the digital output code (displayed by level detectors) specified below:

| $\overline{A0}$ | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| H | L | L | L | L | L | L | L | L | L | L | H |

c. Set switch S3 to position 1 and adjust item P08 until a positive saturation oscilloscope display as defined below is obtained:




Positive saturation oscilloscope display.

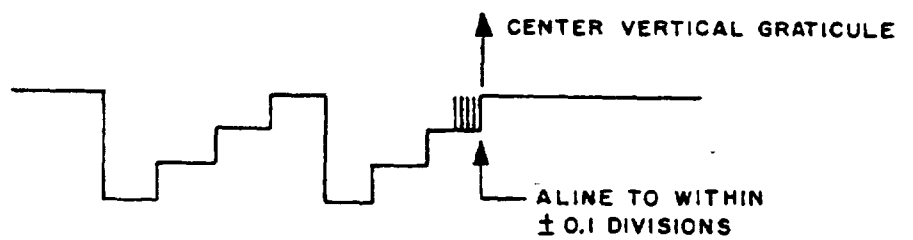
d The setting of item P08 shall satisfy the limits of test conditions Y1 - Y4 of table XXV If the digital output code cannot be obtained, the module shall be considered failed

13. Test conditions Y5 - Y8.

a. Set switches S2 and S3 to position 2 and switches S1 and S4 to position 1.
b Adjust P08 to 9.990 ·0 04 V to obtain the digital output code (displayed by level detectors) specified below

| $\overline{A0}$ | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 | A9 | A10 | A11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| L | H | H | H | H | H | H | H | H | H | H | L |

c Set switch S3 to position 1 and adjust item P08 until a negative saturation oscilloscope display as defined below is obtained.




NEGATIVE SATURATION OSCILLOSCOPE DISPLAY

d. The settings of item P08 shall satisfy the limits of test conditions Y5 - Y8 of table XXV. If the digital output code cannot be obtained, the module shall be considered failed.

FIGURE 38 Zero error and absolute error test circuit for tables XXIV and XXV - Continued



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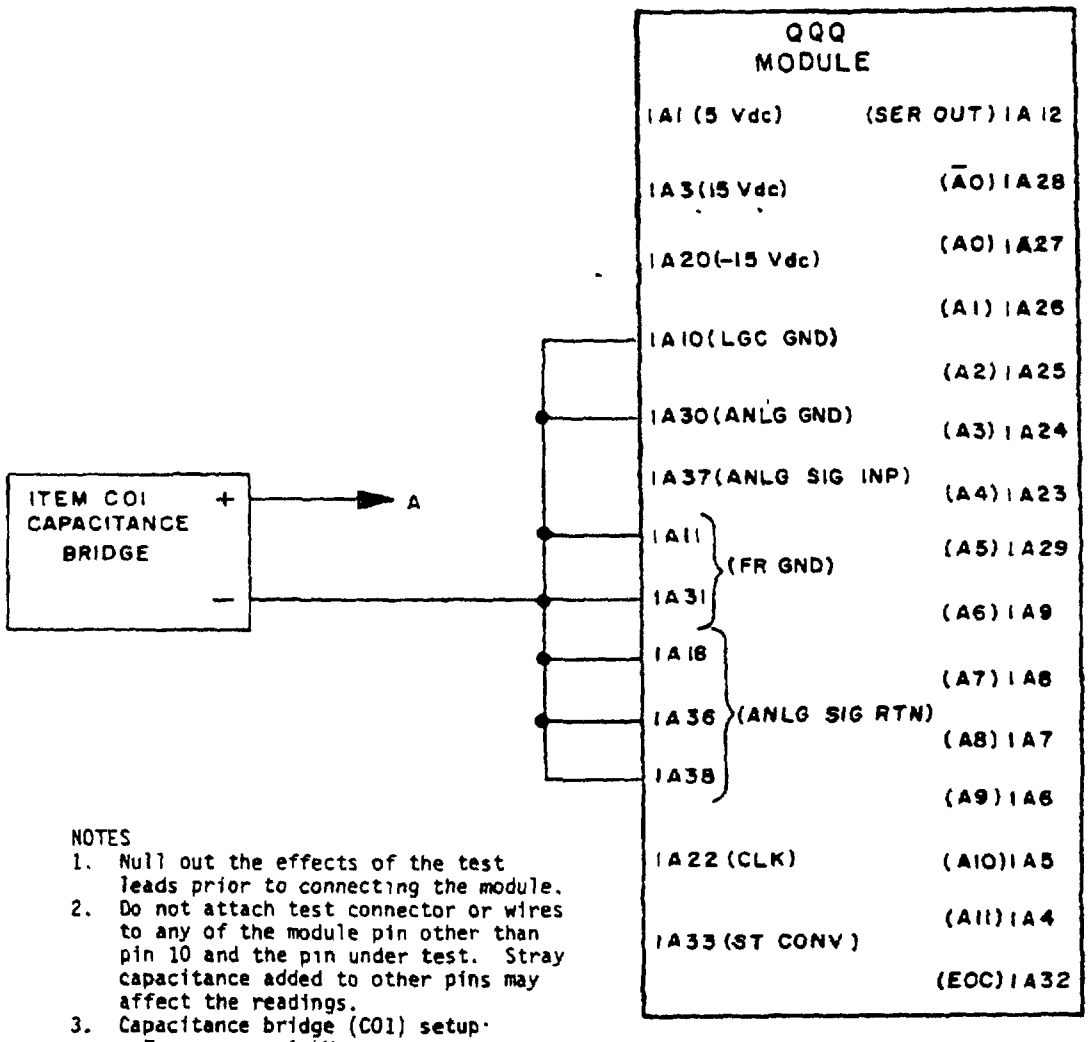
TABLE XXVI. Power supply filter capacitance test.

| Test condition | Parameter | Connect | Initial limits 25° ±5°C Ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| Z1 | Power supply filter capacitance | 1 | 0.08 | 1.50 | µF | 39 |
| Z2 | Power supply filter capacitance | 3 | 0.08 | 1.50 | µF | 39 |
| Z3 | Power supply filter capacitance | 20 | 0.08 | 1.50 | µF | 39 |





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NOTES
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pin other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup:
   Frequency = 1 kHz.
   $V_{PK}$ = 0.3 V maximum.

FIGURE 39 Power supply filter capacitance test circuit for table XXVII.



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